Recent Changes in Firm Dynamics and the Nature of Rising Firm-Size Dispersion*

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Abstract

In Swedish administrative data, I document and decompose a rise in firm-size dispersion over 1997–2017. Rising dispersion among same-aged firms accounts for most of the increase, concentrated among older firms and absent among the young, pointing to forces that accumulate with age. Linked survey data further show that firms of the same age came to differ more in their R&D effort and product scope. In a firm-dynamics model with entry, exit, and permanent productivity differences, a widening productivity gap reproduces this age profile and the aging of the firm population; cheaper expansion or entry barriers do not. The wider gap slows long-run growth but raises the level of productivity through reallocation; accounting for the transition, the welfare cost is only 0.042%, far below what balanced-growth-path comparisons suggest. Consistent with a widening gap, productivity dispersion itself rose in the Swedish data.

Keywords: Firm dynamics, Firm-size dispersion, Administrative data

JEL codes: D22, L11, L25, E23, O47

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1 Introduction

Firm-size dispersion has risen substantially across advanced economies over the past decades (Chen, 2022; Cao et al., 2022; Kwon et al., 2024; Autor et al., 2020).[1] Identifying the channel through which dispersion rises matters because it is informative about the underlying cause: a rise through the aging of the firm population points to different forces than a rise through the divergence of same-aged firms. This paper decomposes the rise in firm-size dispersion into three observable margins of firm dynamics and shows that the overwhelming share of its increase is accounted for by rising heterogeneity among firms of the same age. It then traces this heterogeneity to widening productivity dispersion: a model of firm dynamics in which a productivity gap between firms widens reproduces the documented changes, and productivity dispersion did widen in the same data.

In comprehensive administrative data covering the universe of Swedish firms between 1997 and 2017, cross-sectional firm-size dispersion rose substantially: the interquartile range of log sales widened by 0.5 log points, raising the ratio of the 75th to the 25th size percentile from 7.4 to 12.2. To identify what underlies this rise, I apply the law of total variance, decomposing dispersion in log firm size into three margins of firm dynamics: differences in mean size across firm ages, the firm-age distribution, and size dispersion conditional on age. For the US, Hopenhayn et al. (2022) and Karahan et al. (2024) document two facts about the first two margins: the age distribution has shifted toward older firms, and mean size conditional on age has remained stable. I confirm both in Swedish data — the share of firms younger than two fell from 14.9% to 10.4%, the share aged ten or older rose from 44.9% to 56.4%, and differences in mean size across ages changed little. The third margin is this paper's new fact: size dispersion conditional on age rose substantially among older firms — the interquartile range of log sales widened by about 0.3 log points — while showing no trend among the young. These patterns are robust across sectors and alternative size measures including employment, and they extend to the extreme right tail of the size distribution, where the rise is, if anything, more pronounced. Through the lens of the decomposition, which is model-free and holds exactly in the data, this rising within-age dispersion accounts for 89.2% of the increase in cross-sectional dispersion between 2002 and 2017, while the shift in the age distribution accounts for the remainder and changes in mean size by age contribute negligibly. Despite dominating the rise, size dispersion conditional on age has received comparatively little attention.

What drives the rise in within-age dispersion? Its concentration among older firms is

[1] See also Akcigit and Ates (2021); Grullon et al. (2019); Decker et al. (2016*a*, 2020); Gourio et al. (2014). De Loecker et al. (2020) further document a reallocation of market shares from low- to high-markup firms, consistent with rising firm-size dispersion.

the key discriminating fact: it points to forces whose effect accumulates with firm age, such as productivity divergence (Aghion et al., 2023). Three prominent alternatives — weakening selection among unproductive firms, a changing composition of entrants, and rising measurement error — each imply a footprint across firm ages, sectors, or the tails of the size distribution that the data do not show (Supplemental Appendix B.4). Two firm-level surveys, linked to the same registers, show along which margins same-aged incumbents drifted apart. Among manufacturing firms, the dispersion of the number of products a firm sells rose, whereas the dispersion of sales per product shows no trend. In addition, the dispersion of R&D expenditure and of R&D intensity rose among R&D performers. Firms of the same age thus came to differ more in their innovation effort, which accumulated into increasingly unequal product scope, while the value of their individual products contributed nothing to the divergence.

Motivated by these facts, I ask whether widening productivity dispersion can generate them. Aghion et al. (2023) develop a model of creative destruction with permanent heterogeneity in firm productivity. Their model abstracts from firm entry and exit, leaving mean size by age, the firm-age distribution, and size dispersion conditional on age undefined. I embed this mechanism — permanent productivity heterogeneity — into a quality-ladder model of firm dynamics with entry and exit in the spirit of Klette and Kortum (2004), in which these objects are well defined. Firm size has the two margins of the survey evidence: the number of products a firm sells, which responds to productivity differences, and the sales per product, which vary with product-line weights that I hold fixed, since their dispersion did not change over the period. The weights are neutral for firm dynamics and aggregate outcomes. I calibrate the model to firm dynamics around the turn of the millennium, with the dispersion of the weights set to match the level of size dispersion.

I then ask how much of the subsequent changes a single force — a widening productivity gap — can explain: holding all other parameters at their baseline values, I raise the productivity gap to match the rise in size dispersion conditional on age among older firms and the 4.5 percentage-point fall in the share of young firms. A modest rise reproduces the dispersion target and most of the fall in the young-firm share, despite the fit being overidentified. The mechanism is differential expansion: more productive firms expand faster, so size differences cumulate with tenure, raising within-age dispersion among older firms while leaving young-firm dispersion flat — the age signature in the data. More dispersed weights, by contrast, would raise dispersion at every age, one for one among entrants, where the data are flat. Without being targeted, the wider gap also raises the share of firms aged ten or older by 11 percentage points, close to the 12-point rise in the data, and slows long-run productivity growth by 0.10 percentage points relative to its baseline of 1.5%. The source of the slowdown is entry: as product lines shift toward the more productive type, the

expected profits of an entrant's product fall, and so does the flow of entrants.

The age signature points to a gap between persistent firm types. As a robustness check, I fit a lower cost of firm expansion or a higher cost of entry — natural alternatives from the literature on declining business dynamism — to the same two targets. Neither reproduces the age signature: both raise dispersion among firms of all ages and steepen the mean size–age profile, counter to the data, and both understate the widening of the far right tail among old firms. A widening gap in R&D efficiency between the two types reproduces the age signature as well, since it works through the same divergence in expansion rates. It falls short, however, on the young-firm share and, unlike the productivity gap, raises aggregate growth, as does a cheaper expansion technology. The fall in the young-firm share, the observed productivity slowdown, and the direct evidence on productivity dispersion presented below thus point to the productivity gap.

The growth slowdown is central to the model's welfare implications, which I evaluate by solving the full transition between balanced growth paths. The wider gap reallocates market share toward more productive firms, raising the level of aggregate productivity, but slower long-run growth is costly to consumers. Along the transition, output enjoys a modest, temporary boom before slower growth pulls it below its previous trend. Accounting for the transition, the level gain offsets most of the growth loss, leaving a welfare cost of only 0.042% in consumption-equivalent terms — far below the 1.92% implied by comparing balanced growth paths alone, which imposes the slowdown at once and misses the level gain from reallocation.

The model's exogenous shifter can itself be verified in the data. Revenue productivity, which balance-sheet data observe for firms of all sectors, sizes, and ages, became more dispersed within industries over the period, among young firms as much as among all firms, and so did physical productivity among surveyed manufacturers, for which prices are observed. Firms of the same age thus became more dispersed in productivity already when young, whereas the size differences this generates appear only with tenure — the signature of the mechanism.

Further literature. This paper contributes to a literature on the drivers of recent macroeconomic trends — rising firm-size dispersion, declining business dynamism, and slowing productivity growth (Aghion et al., 2023; De Ridder, 2024; Akcigit and Ates, 2023; Liu et al., 2022; Peters and Walsh, 2021; Paron, 2026). The decomposition supplies disciplining moments for this literature: any candidate mechanism must generate rising within-age dispersion among incumbents, concentrated at older ages.

A separate literature uses models of firm dynamics to explain the firm-size distribution itself, and in particular its heavy, approximately Pareto tail, with dispersion arising

from idiosyncratic growth, selection, and entry (Klette and Kortum, 2004; Luttmer, 2007; Gabaix, 2009). That work concerns the *level* of firm-size dispersion. This paper concerns instead its *change*: how dispersion has risen over time, and which margins of firm dynamics account for that rise. The model accordingly takes the level as given. It loads the level on the dispersion of the product-line weights rather than on productivity differences, in line with the finding of Hottman et al. (2016) that differences in firm appeal, rather than in costs, account for most of the cross-sectional variance of firm sales.[2]

The findings also relate to Decker et al. (2016*b*), who document a decline in the dispersion of firm *growth rates*. Rising dispersion in size *levels* conditional on age is consistent with their finding if growth has become more persistent and type-driven — the same firms increasingly experiencing high or low growth. Decker et al. (2020) provide direct evidence that firm growth has become less responsive to shocks, supporting this interpretation.[3]

Finally, the widening dispersion in firm productivity that drives the model is documented not only in the Swedish data of Section 7 but also in the literature. Andrews et al. (2016) show that both labor productivity and multifactor productivity diverged between the most productive firms, the global frontier, and the rest throughout the 2000s, a gap robust to controls for markups. Berlingieri et al. (2017) document the same divergence in labor and multifactor productivity within many OECD countries, and link the growing gap between high- and low-productivity firms to digitalization.

The remainder of the paper is organized as follows. Section 2 introduces a model-free decomposition of firm-size dispersion based on the law of total variance. Section 3 describes the administrative data. Section 4 documents trends in firm-size dispersion, decomposes its rise, and traces the divergence among incumbents to product scope and R&D in linked survey data. Section 5 develops a model of firm dynamics with permanent productivity heterogeneity, and Section 6 calibrates it, raises the productivity gap to match the documented changes, tests it against alternative shifters, and computes the transition and welfare effects. Section 7 confronts the mechanism with direct evidence on productivity dispersion. Section 8 concludes.

[2]The Swedish survey data point the same way: sales per product are considerably more dispersed across firms than product counts.

[3]In the model in this paper, rising dispersion in permanent firm productivity raises the dispersion of the type-specific, persistent component of firm growth, but at the same time reduces dispersion in the idiosyncratic component: creative destruction falls, so product loss becomes rarer, and the economy is increasingly populated by large firms operating many products, for which the loss of a single product is proportionally smaller and growth correspondingly less random. Other forces, such as rising labor adjustment costs (Decker et al., 2020), could also account for the decline in the dispersion of growth rates.

2 A decomposition of firm-size dispersion

To structure the empirical analysis, I begin with a model-free decomposition of firm-size dispersion. Consider the law of total variance, $\text{Var}(Y) = \text{Var}(\mathbb{E}[Y|X]) + \mathbb{E}[\text{Var}(Y|X)]$, which decomposes the total dispersion in Y into (i) variation across groups defined by X and (ii) variation within those groups. Setting $Y \equiv \ln s_f$ and $X \equiv a_f$, where s_f and a_f denote firm size and age, respectively, this decomposition partitions cross-sectional dispersion in log firm size into two components: differences in average size across firm ages and size dispersion conditional on age. Specifically,

$$\text{Var}\left(\ln s_f\right) = \sum_{a_f} p(a_f) \left[\mathbb{E}[\ln s_f | a_f] - \mathbb{E}[\ln s_f]\right]^2 + \sum_{a_f} p(a_f) \text{Var}(\ln s_f | a_f), \tag{1}$$

where $p(a_f)$ is the share of firms of age a_f and $\mathbb{E}[\ln s_f] = \sum_{a_f} p(a_f)\mathbb{E}[\ln s_f | a_f]$ the average log size of all firms. The definition of firm age can be either discrete (e.g., integer years) or based on broader age bins; the decomposition applies in either case.

Although it is stated in terms of the variance, the decomposition more generally reveals that cross-sectional firm-size dispersion is shaped by three margins: (i) differences in mean size across firm ages, $\mathbb{E}[\ln s_f | a_f]$; (ii) dispersion in size within age groups, $\text{Var}(\ln s_f | a_f)$; and (iii) the age distribution, $p(a_f)$. Accordingly, total dispersion can increase for three distinct reasons: greater divergence in mean size across ages, increased within-age heterogeneity, or shifts in the age distribution toward older firms, which exhibit greater within-age dispersion.[4]

How have these margins evolved over time? I first describe the data and then examine each margin in turn.

3 Data

The data come from Statistics Sweden's main firm-level dataset, Företagens Ekonomi, which contains annual balance sheet and income statement information for the universe of Swedish firms over the period 1997–2017. It includes firm-level variables such as sales, assets, intermediate inputs, and employment, as well as information on industry and legal form. I restrict the sample to firms in the private sector that eventually employ at least one worker. For a detailed description of the data, see Section A in the Supplemental Appendix.

[4]The focus of this paper is on decomposing *changes* in firm size dispersion over time. See Kondziella (2026) for a decomposition of its *level* into size differences across firm ages and dispersion conditional on age using eq. (1).

Firm age is defined as the number of years since the first worker — either employee or self-employed — joins the firm. This information comes from the auxiliary dataset Registerbaserad Arbetsmarknadsstatistik, which covers the universe of employer-employee matches and extends back to 1992. This allows me to observe untruncated firm age for all firms founded from 1993 onward.

For the discussion in Section 4.4, I link two further surveys by Statistics Sweden to the firm-level data: the survey of industrial production (Industrins varuproduktion, IVP), which records each manufacturing firm's production by product code, and the business R&D survey (Forskning och utveckling i företagssektorn, FoU). Both are described in Supplemental Appendix A.

All nominal variables are deflated to 2017 Swedish kronor (SEK) using the GDP deflator. The results are virtually unchanged when using sector-specific deflators; moreover, sectoral deflation does not affect measures of within-sector dispersion in log firm size.

Table 1: Summary statistics

	25th Pct.	50th Pct.	75th Pct.	Mean	SD
*Sales**	0.4	1.4	4.3	10.9	62.1
*Intermediate Inputs**	0.2	0.6	1.8	4.9	34.8
*Capital stock**	0.0	0.2	0.8	4.6	96.2
Employment	0	1	3	4.9	29.3
*Wage bill**	0.0	0.2	0.9	1.8	11.0

Notes: variables marked with * are in units of million 2017-SEK (1 SEK $\approx$ 0.1 US dollars). The capital stock is defined as fixed assets minus depreciation. Statistics are computed on the pooled sample of firm-years over 1997–2017, comprising 7,590,605 observations.

The final dataset comprises 7,590,605 firm-year observations; Table 1 reports summary statistics for the pooled sample. The median firm has sales of approximately 1.4 million SEK (1 SEK $\approx$ 0.1 USD) and employs one full-time equivalent worker. The distributions of sales and inputs are highly right-skewed: mean sales (10.9 million SEK) and employment (4.9 workers) substantially exceed their respective 75th percentiles. A nontrivial fraction of firms report zero capital stock (fixed assets net of depreciation), employment, or wage bill, as reflected in zero-valued 25th percentiles for these variables. Zero employment is common among the self-employed, which is typical in the early years of a firm's life, before any additional workers are hired. This feature has implications for the choice of the firm-size measure.

I use firm sales as the baseline measure of size and provide robustness checks using

alternative proxies.[5] Sales offer two advantages. First, few firms report zero sales, which avoids the log-of-zero problem that arises with employment. Second, sales provide a more continuous measure of firm size, whereas employment is highly discrete. For example, the 25th, 50th, and 75th percentiles of employment are 0, 1, and 3, respectively, limiting the informativeness of dispersion measures such as the interquartile range (IQR). I measure size dispersion by the IQR in Figures 1 and 4 and by the standard deviation in the decomposition, since eq. (1) is stated for the variance. To mitigate the influence of outliers, sales are winsorized at the 0.1 and 99.9 percent levels.[6]

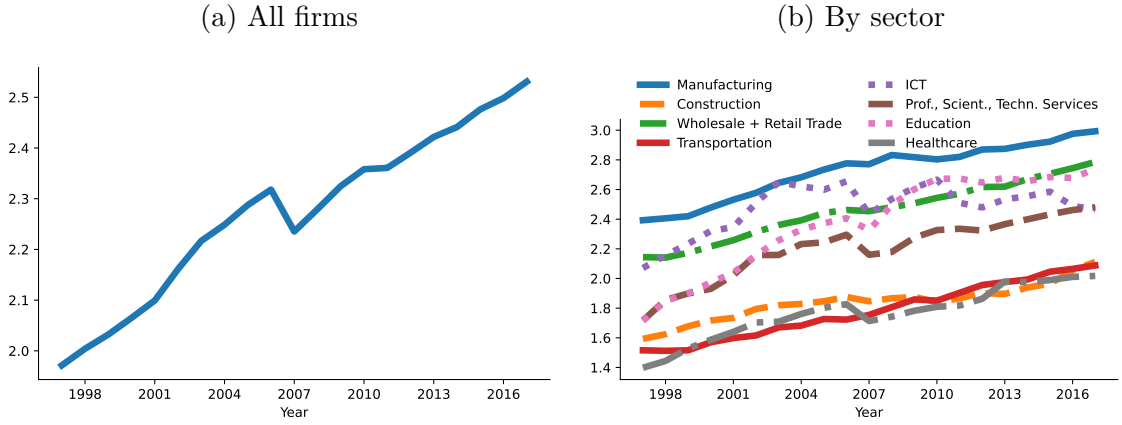
4 Size dispersion and firm dynamics in the data

This section examines cross-sectional size dispersion and the three margins of the decomposition in eq. (1) over time. All margins are documented in the same data, allowing for an exact decomposition of firm-size dispersion.

4.1 Firm-size dispersion

Figure 1 shows cross-sectional firm-size dispersion over time, measured by the interquartile range (IQR) of log sales. The IQR rises from 2.0 to 2.5 log points, implying that the 75th-to-25th percentile size ratio grew from 7.4 to 12.2.

Figure 1: Cross-sectional firm-size dispersion



Notes: The figure shows the interquartile range (IQR) defined as the difference between the 75th and 25th percentile of the log sales distribution. The left panel shows the IQR for all firms; the right panel shows the IQR by sector.

[5] Throughout, sales are interpreted as a proxy for firm size rather than market share.

[6] The filter is applied on the pooled sample after deflating.

The right panel shows the same dispersion measure by sector. The increase in dispersion is shared across all sectors and is of similar magnitude in each, with most displaying the same 0.5 log point rise shown in the left panel. Hence, the rise in aggregate dispersion is not driven by differential trends across sectors.

Unlike the standard deviation, the IQR does not require dropping firms with zero sales, since the 25th percentile of the sales distribution is strictly positive. Beyond this practical advantage, the IQR is robust to the extreme right-skewness of the sales distribution and captures dispersion in the bulk of the firm population rather than being driven by the tails. The picture is virtually unchanged when dispersion is measured by the standard deviation instead, dropping firms with zero sales (Figure A-1, Supplemental Appendix): the standard deviation increases by 0.4 log points, close to the 0.5 log point increase in the IQR. The extreme right tail is documented directly, using unwinsorized sales (Figure A-9, Supplemental Appendix): the gap between the 99th and the 25th percentile of log sales widens by 0.7 log points, so the rise in dispersion extends to the extreme right tail and is more pronounced there.

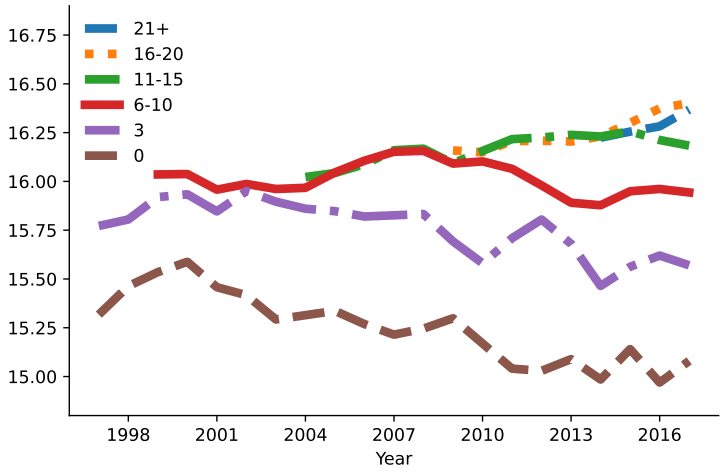
4.2 Firm dynamics

Firm size conditional on age Figure 2 shows the log of the average firm size by firm age.[7] Two patterns are noteworthy. First, there are stark level differences across ages: older firms are, on average, much larger than younger firms. At the start of the sample, the average firm size among firms aged 6–10 is 0.501 log points higher than among entrants (age zero). Second, mean size conditional on age is broadly stable over time. Firms of older ages show no systematic trend over time. Young firms, represented in the graph by firms aged zero or three, show a slightly negative trend in average sales toward the end of the sample, which is not visible in employment (Figure A-2, Supplemental Appendix). Similar patterns were first documented by Hopenhayn et al. (2022) and Karahan et al. (2024) for the US and by Engbom (2023) for Sweden.

Firm-age distribution Figure 3 shows two margins of the firm-age distribution: the young-firm share in the left panel and the share of mature firms in the right panel. The young-firm share, defined as the share of firms aged zero (entrants) or one, declined from 14.9% in 1997 to 10.4% in 2017. The share of mature firms, defined as firms at least ten years old, increased from 44.9% in 2002, the first year in which this group can be identified, to 56.4% in 2017, an increase of about 12 percentage points over 15 years. Hopenhayn et al. (2022) and Karahan et al. (2024) previously

[7] Note that the decomposition in eq. (1) contains the average of the log firm size rather than the log of the average firm size. I show the latter to avoid dropping firms with zero sales.

Figure 2: Firm size conditional on age



Notes: The figure shows the log of the average firm size (sales) by firm age indicated in the legend. Sales are in 2017 SEK.

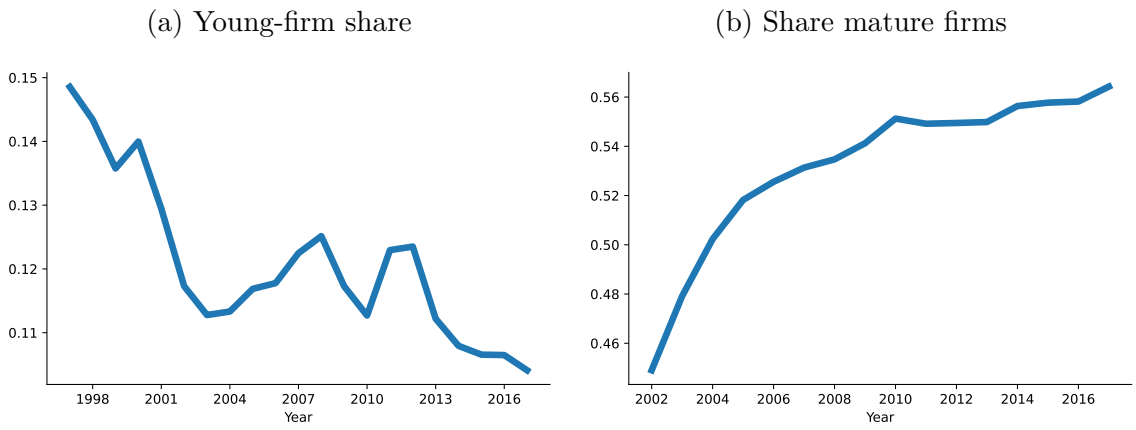
documented a quantitatively similar shift in the firm-age distribution for the US.

Firm-size dispersion conditional on age The left panel of Figure 4 shows the third and last element of the dispersion decomposition in eq. (1), namely size dispersion conditional on age, again measured using the IQR of log sales. Two points are noteworthy. First, there are stark differences across ages: among older firms, sizes are much more dispersed than among younger firms. Second, dispersion has risen systematically over time, but the increase is concentrated among older firms. For ages 0 and 3, size dispersion shows no trend over the sample period. For age groups 6–10, 11–15, and 16–20, within-group size dispersion increased by about 0.3 log points.

The right panel of Figure 4 shows size dispersion within the age group 11–15 by sector, with all sectors normalized to zero at the start. All sectors show an increase in size dispersion conditional on age. The increase is quantitatively similar across sectors, with most displaying a rise of about 0.3 log points consistent with the rise shown in the left panel for this age group. Among the sectors, Manufacturing displays the smallest increase whereas Healthcare shows the largest. Hence, the rise in firm-size dispersion conditional on age is a within-sector phenomenon.

These trends are robust to a range of specification choices, as shown in Supplemental Appendix B: measuring dispersion by the standard deviation of log sales (excluding firms with zero sales), conditioning on integer ages without any age groupings, deflating firm sales with sector-level GDP deflators, excluding mergers and acquisitions,

Figure 3: Firm-age distribution



Notes: The left panel shows the young-firm share, defined as the share of firms aged zero or one. The right panel shows the share of firms that are at least ten years old.

and measuring firm size by intermediate inputs, capital, or employment. Figure A-9 in the same appendix further documents the extreme right tail of the size distribution directly, replacing the 75th by the 99th percentile and using unwinsorized sales: the rise in dispersion among older firms is also present in the right tail and is, if anything, more pronounced there.

4.3 Decomposing the rise in firm-size dispersion

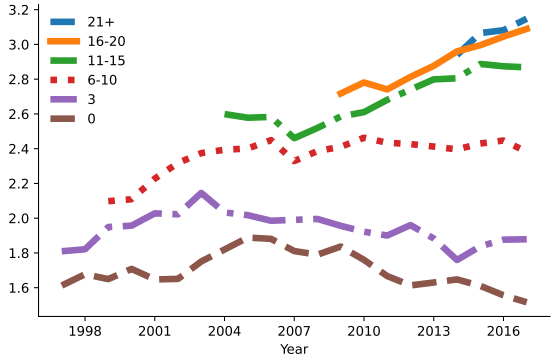
The graphical analysis in the previous subsection documented a shift in the firm-age distribution toward older firms and a rise in size dispersion conditional on age. How much does each margin contribute quantitatively to the rise in cross-sectional size dispersion?

To answer this question, I fix two points in time, compute the cross-sectional dispersion of firm size using eq. (1), and then separately change either the average size conditional on age, the firm-age distribution, or size dispersion conditional on age. I do so twice, once starting from the initial point and changing each component to its value in the end period and once vice versa, and report the average of both. Because the decomposition is nonlinear, the contributions of the three margins do not necessarily sum to the total change, but in practice they do so up to the fourth decimal. For this decomposition, I restrict to firms with positive sales so that I can implement the decomposition in eq. (1) for the standard deviation exactly.

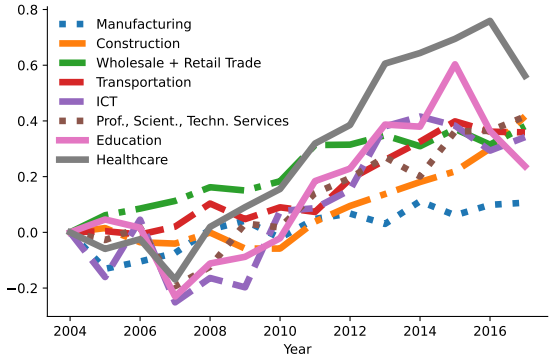
The end point is the last year in the data, 2017. For the start point, an earlier year lengthens the period that can be decomposed, but firm age is untruncated only

Figure 4: Firm-size dispersion conditional on age

(a) All firms



(b) By sector (ages 11–15)



Notes: The figure shows the interquartile range (IQR) defined as the difference between the 75th and 25th percentile of the log sales distribution. The left panel shows the IQR for all firms conditional on age as indicated in the legend; the right panel shows the IQR by sector for firms aged 11–15.

for firms born 1993 or later, so an earlier start requires a coarser age grouping for older firms. I choose 2002, which allows me to decompose a period of 15 years and to differentiate firm ages up to age ten, i.e., in the decomposition in eq. (1), $a_f \in \{0, 1, \ldots, 8, 9, 10+\}$. Note that the decomposition in eq. (1) holds for any age grouping.

Between 2002 and 2017, dispersion of firm sales increased by 0.202 log points, as shown for the standard deviation in Figure A-1 (Supplemental Appendix). Table 2 decomposes this rise into its components. Changes in mean size conditional on age contributed −0.006 log points, or −2.8% of the total, and thus leave no mark on the rise in dispersion. The shift in the age distribution contributed positively but modestly, at 0.027 log points (13.6% of the total). Lastly, the increase in size dispersion conditional on age contributed 0.180 log points (89.2% of the total). Rising within-age heterogeneity thus accounts for the overwhelming majority of the increase in size dispersion. Table 2 further shows that this result is robust across sectors. In most sectors, changes in mean size conditional on age make a small negative contribution, while changes in the age distribution contribute modestly and positively. By contrast, changes in size dispersion conditional on age are large and positive.[8]

[8]The ICT sector, where the largest part is accounted for by the aging of firms, is the only exception. This is specific to the decomposed time period: the decomposition starts in 2002, shortly after the dotcom boom, which saw an increase in entrant activity in the ICT sector. The ICT sector was therefore characterized by a relatively young firm population at that time, which experienced substantial aging over the subsequent period.

Table 2: Decomposition of the change in $\sqrt{\text{Var}(\ln s_f)}$ (2002–2017)

	$\mathbb{E}[\ln s_f \mid a_f]$	$p(a_f)$	$\text{Var}(\ln s_f \mid a_f)$	Total
Total economy	-0.006 (-2.8)	0.027 (13.6)	0.180 (89.2)	0.202
Manufacturing	-0.016 (-6.3)	0.052 (20.0)	0.226 (86.3)	0.262
Construction	-0.004 (-1.4)	-0.006 (-2.3)	0.275 (103.8)	0.265
Wholesale + Retail Trade	-0.009 (-3.0)	0.031 (10.4)	0.271 (92.6)	0.293
Transportation	-0.007 (-2.4)	0.028 (9.3)	0.276 (93.1)	0.297
ICT	-0.008 (-5.1)	0.096 (64.7)	0.060 (40.5)	0.149
Prof., Scient., Techn. Services	0.008 (4.2)	0.023 (12.0)	0.158 (83.7)	0.189
Education	0.001 (0.5)	0.050 (18.8)	0.215 (80.7)	0.266
Healthcare	0.009 (4.5)	-0.008 (-4.2)	0.187 (99.7)	0.187

Notes: The table decomposes the change in the standard deviation of log firm sales, $\sqrt{\text{Var}(\ln s_f)}$, between 2002 and 2017. Each component is computed by varying one element at a time — the conditional mean $\mathbb{E}[\ln s_f \mid a_f]$, the age distribution $p(a_f)$, and within-age dispersion $\text{Var}(\ln s_f \mid a_f)$ — while holding the others fixed. Each element is varied relative to the start and end period and the table reports the average of both. Each cell reports the contribution in log points and, in parentheses, as a percentage of the total change; because the components do not sum exactly to the total and the entries are rounded, the percentages need not sum to 100.

4.4 Product scope and R&D

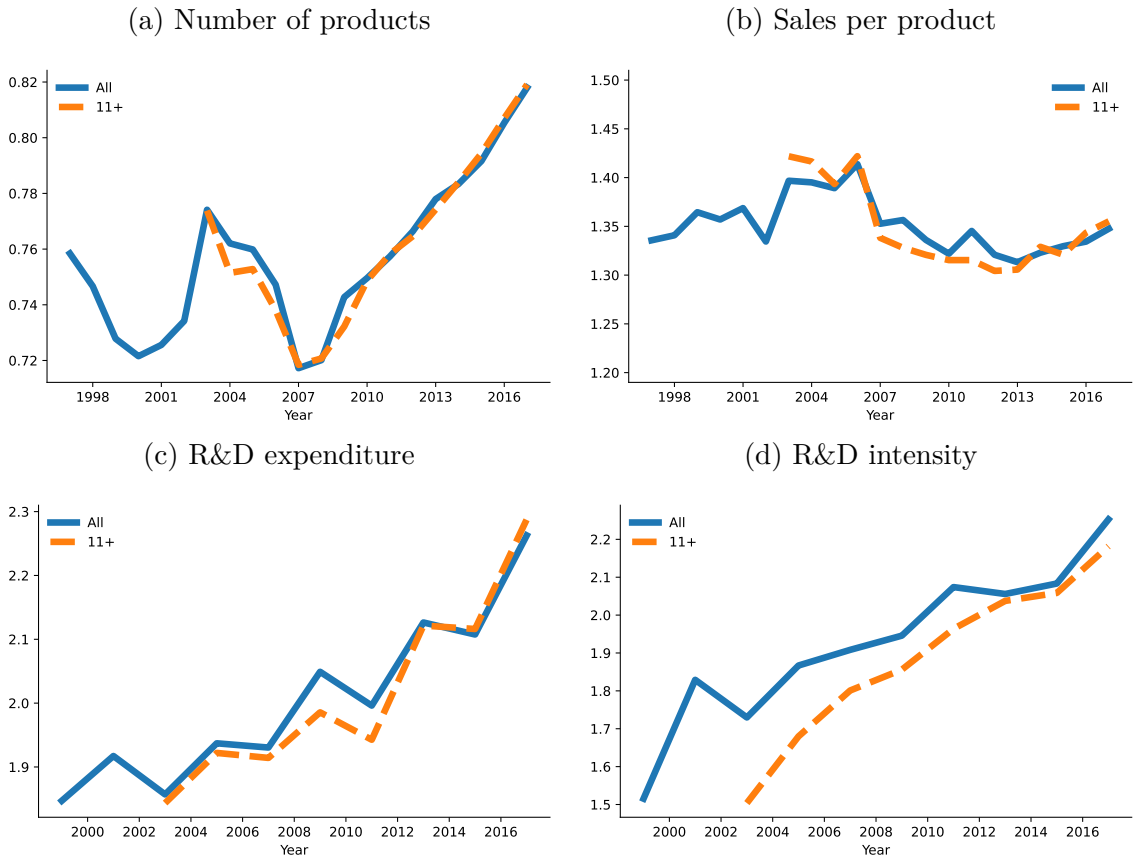
The dominant role of within-age heterogeneity disciplines theories of the rise in firm-size dispersion. Any such theory must answer a single question: what drives rising within-age dispersion? The age profile of the rise says that incumbents of the same age drifted apart. Supplemental Appendix B.4 rules out three mechanisms — weakening selection among unproductive firms, a changing composition of entrants, and rising measurement error — each of which implies a footprint across firm ages, sectors, or the tails of the size distribution that the data do not show. Here I turn to two firm-level surveys by Statistics Sweden that observe, for the same firms, the margins along which incumbents can diverge: the number of products they sell and the amount they spend on R&D.

The survey of industrial production (IVP) records, for every manufacturing firm above a size cut-off, the value of production by product code — the eight-digit Combined Nomenclature, refined by Statistics Sweden. A firm's count of product codes is as close to the notion of a product in a multi-product firm model as observable data get.[9] The business R&D survey (FoU) records, every second year, a firm's expenditure on R&D performed in-house. A distinctive advantage of the Swedish registers is that both surveys can be linked at the firm level to the balance-sheet data used so far,

[9] Berlingieri et al. (2024) use the French counterpart of these data and show that firms add products in bursts.

so the size trends, the product counts, and the R&D of the same firms are observed together. Supplemental Appendix A describes both surveys and the samples. In short, I keep surveyed manufacturing firms with at least 20 employees and count their own-produced goods at the six-digit level, and I keep R&D survey respondents with at least 50 employees — the size class inside the survey's frame in every wave — that report positive own R&D. Both samples are therefore tilted toward larger and, in the case of IVP, manufacturing firms; recall that Manufacturing is the sector with the smallest rise in size dispersion conditional on age (Figure 4), so the product evidence is, if anything, a lower bound.

Figure 5: Dispersion of product scope and R&D among incumbents



Notes: Each panel shows the standard deviation of a log firm-level variable across firms, for all firms (solid) and for firms aged 11 and older (dashed; from 2003, the first year in which this group is fully identified). Panels (a) and (b) use the survey of industrial production: surveyed manufacturing firms with at least 20 employees, own-produced goods; the number of products is the number of distinct six-digit product codes with positive production value, and sales per product is the firm's production value divided by that number. Panels (c) and (d) use the R&D survey: respondents with at least 50 employees reporting positive own R&D expenditure; R&D intensity is own R&D expenditure divided by net sales. Firms younger than 11 are not drawn separately because the survey cells are thin; the young-firm evidence rests on Figure 4. Details in Supplemental Appendix A.

Figure 5 reports the results. Panel (a) shows the dispersion of the number of products. For all surveyed firms it rises from 0.76 log points in 1997 to 0.82 in 2017. The rise holds conditional on age: among firms aged 11 and older, the one age group with enough surveyed firms to be shown separately, dispersion rises from 0.77 in 2003 to 0.82 in 2017, with the increase concentrated after 2007, when the series stands at 0.72. Panel (b) shows the other component of a firm's sales, the value of sales per product. Its dispersion shows no systematic trend: 1.34 log points in 1997, 1.35 in 2017, and between 1.31 and 1.41 in every year in between. Of the two components of a firm's sales, the one along which surveyed manufacturing firms drifted apart is therefore the number of products a firm sells, not how much each product sells.

Panels (c) and (d) turn to the input side. The dispersion of R&D expenditure across firms that perform R&D rises from 1.85 log points in 1999 to 2.26 in 2017, and from 1.84 in 2003 to 2.29 among firms aged 11 and older. R&D intensity — R&D per unit of sales — disperses even more, from 1.52 to 2.25 for all performers and from 1.50 to 2.18 among the older ones: relative to their sales, firms of the same age spend increasingly different amounts on innovation.

Read together, the four panels tell a two-step story that carries over to the model. The diverging R&D expenditures and intensities show that firms of the same age came to differ more and more in their innovation effort. The diverging product counts show what this effort accumulates into: a wider spread in the number of products firms sell. The constant dispersion of sales per product over 1997–2017 adds that the value of a firm's individual products contributed nothing to this divergence. The model below can therefore treat that value layer as fixed and let the entire rise in size dispersion run through the number of products. The linked data also let me confirm, in this sample, the core relationships between firm size, product scope, and R&D on which such a model rests (Table A-1 in Supplemental Appendix A).

5 Model

The evidence of Section 4 sets three tasks for a theory of the rise in firm-size dispersion. It must raise size dispersion among older firms while leaving dispersion among the young and the mean size–age profile unchanged; it must shift the age distribution toward older firms; and it must work through the margins along which incumbents were seen to diverge, with increasingly unequal innovation effort accumulating into increasingly unequal product scope while the value of individual products contributed nothing. A model of firm dynamics in which firms grow by adding product lines through R&D, in the spirit of Klette and Kortum (2004), has these objects built in: R&D determines the rate at which a firm adds products, entry and the loss of

products to competitors determine survival and the age distribution, and firm size is the sum of the products a firm holds.

What the model lacks is a reason for firms of the same age to drift apart. I add one: permanent differences in productivity across firms, the force that Aghion et al. (2023) study in a model without entry and exit and that Section 7 documents directly in the Swedish data. A more productive firm earns a higher markup on the product lines it takes from less productive rivals, so expansion is worth more to it and it expands faster; its size pulls away from that of an equally old but less productive firm as tenure lengthens. Firm size has two margins in the model, mirroring Section 4.4: the number of products a firm sells, which responds to productivity differences, and the sales per product, which vary with product-line weights that I hold fixed. The remaining building blocks are standard. I describe them briefly, relegate derivations to Supplemental Appendix D, and concentrate on what the productivity types and the product-line weights add.

5.1 Environment

Time is continuous. A representative household with discount rate ρ supplies one unit of labor inelastically and chooses consumption C_t to maximize $\int_0^\infty \exp(-\rho t) \ln C_t \, dt$, so that consumption growth satisfies the Euler equation $\dot{C}_t/C_t = r_t - \rho$, where r_t denotes the interest rate. The final good is produced competitively from a continuum of products, indexed by i, with a Cobb-Douglas technology (t subscripts are suppressed when convenient),

$$Y = \exp\left(\int_0^1 \beta_i \ln \left[\frac{q_i y_i}{\beta_i} \right] di \right),$$

where y_i and q_i denote the quantity and quality of product i, and β_i is the weight of product i in final output. Output is consumed entirely, $Y = C$, and its price is normalized to one. Expenditure on product i equals $p_i y_i = \beta_i Y$, so that β_i is the share of aggregate expenditure that falls on product i.

The product-line weights capture the fact documented in Section 4.4 that sales per product differ widely across firms, with a dispersion that shows no trend over time.[10]

[10] In Kondziella (2026), I model the sales-per-product margin explicitly. There, the final good aggregates product lines with an elasticity of substitution above one, so that a line's sales rise with its quality and the dispersion of sales per product is an equilibrium object, shaped by innovation and by firm types, rather than a distribution of fixed weights. That paper explains the level of dispersion. This paper explains its rise, and the dispersion of sales per product shows no trend over the period (Figure 5, panel b), so I take this margin as exogenous here, in the form of a fixed distribution of weights.

I make three assumptions. First, the weights are fixed over time and drawn independently across products from a common distribution G with mean one, so that $\int_0^1 \beta_i di = 1$. Second, innovation is undirected, as in Klette and Kortum (2004): firms cannot target products with a high weight. Third, at time zero the weight of a product is independent of its quality and of the productivity types of the firms competing for it. The second assumption preserves this independence at all later dates. The standard formulation with equal weights is the special case $\beta_i = 1$ for all i. As I show below, the weights shape the cross-sectional distribution of firm sales and nothing else: firm policies and all aggregate outcomes are the same for any distribution G.

Firms, indexed by f, hire labor at the wage w and produce with a linear technology, $y_{if} = \varphi_f l_{if}$, where l_{if} denotes the labor employed in product line i and φ_f the productivity of the firm. Productivity is time-invariant and common to all product lines of a firm: some firms are systematically more efficient than others. As in Aghion et al. (2023), there are two productivity types, $\varphi_f \in \{\varphi^h, \varphi^\ell\}$ with $\varphi^h/\varphi^\ell > 1$, which I refer to as high- and low-type firms.[11]

5.2 Prices, firm size, and aggregate output

Prices are set as in Aghion et al. (2023). Firms compete in prices within each product line, so that product i is supplied by the firm with the highest quality-adjusted productivity $q_{if}\varphi_f$, the incumbent, which sets its price equal to the quality-adjusted marginal cost of the follower f', the firm with the second-highest quality-adjusted productivity. The markup of the incumbent is therefore the product of its quality and productivity advantages over the follower,

$$\mu_{if} \equiv \frac{p_{if}}{w/\varphi_f} = \frac{q_{if}}{q_{if'}} \frac{\varphi_f}{\varphi_{f'}}. \tag{2}$$

Sales of product i equal $\beta_i Y$, profits $\beta_i Y \left(1 - \mu_{if}^{-1}\right)$, and employment $\beta_i \frac{Y}{w} \mu_{if}^{-1}$. All three scale with the weight of the product, whereas the markup does not depend on it.

Let N_f denote the set of product lines in which firm f is the incumbent. Firm sales

[11] Alternative mechanisms work similarly; in De Ridder (2024), for instance, ex-ante heterogeneity in intangible capital adoption, which lowers the marginal costs of some firms, is akin to ex-ante heterogeneity in productivity.

are

$$\text{Size}_f = Y \sum_{i \in N_f} \beta_i. \tag{3}$$

A firm's sales thus reflect two margins: the number of products it sells, $n_f \equiv \left|N_f\right|$, and the weights of these products. Since the weights have mean one, the expected sales of a firm with n_f products are $n_f Y$; with equal weights, sales are exactly proportional to the number of products. Firm employment, $\frac{Y}{w} \sum_{i \in N_f} \beta_i \mu_{if}^{-1}$, in addition decreases in the markups the firm charges.

In the aggregate, the weights play no role. Because they have mean one and are independent of markups across products, an expenditure-weighted average across products, $\int_0^1 \beta_i h_i di$, of any product characteristic h_i that is independent of the weights equals the unweighted average $\int_0^1 h_i di$. The weights thus drop out of all aggregates, and I omit them in the expressions that follow. Production labor is $L_P = \frac{Y}{w} \int_0^1 \mu_i^{-1} di$, and aggregate output is

$$Y = Q \Phi \mathcal{M} L_P, \tag{4}$$

where

$$Q = \exp\left(\int_0^1 \ln q_i di\right), \quad \Phi = \exp\left(\int_0^1 \ln \varphi_{f(i)} di\right), \quad \mathcal{M} = \frac{\exp\left(\int_0^1 \ln \mu_i^{-1} di\right)}{\int_0^1 \mu_i^{-1} di}.$$

Aggregate total factor productivity is the product of three terms: the geometric averages of product quality, Q, and of the productivity of the incumbent producers, Φ, and a misallocation term, $\mathcal{M} \leq 1$, that reflects the dispersion of markups across products. The term Φ is specific to an economy with productivity types: it rises when product lines are reallocated toward high-type firms. Supplemental Appendix D.1 contains the derivations.

5.3 Innovation and entry

Firms grow through expansion R&D, as in Klette and Kortum (2004). An incumbent chooses a rate of expansion R&D, x_{it}, for each of its product lines i at a cost of x_{it}^{ζ}/ψ_x units of labor, where ψ_x denotes the efficiency of expansion R&D and $\zeta > 1$ the curvature of its cost. Costs are additive across product lines, which yields a closed-form value function. At the rate x_{it}, the firm improves the quality of a randomly selected product of a competitor by a factor of λ and becomes its incumbent

producer.[12] Potential entrants produce a flow rate of entry z_t with a technology that is linear in labor, $z_t = \psi_z L_{Et}$, where ψ_z denotes the entry efficiency and L_{Et} the research labor of entrants. An entrant likewise improves the quality of a randomly selected product by one step and becomes its incumbent. Its productivity type is realized after entry: high with probability p^h and low with probability $1 - p^h$. Every product is therefore lost to a competitor at the rate of creative destruction

$$\tau_t = \int_0^1 x_{it} di + z_t. \tag{5}$$

Permanent productivity differences enter the problem of the firm through the markups on the product lines it captures. Every product line carries a quality gap of one step. By eq. (2), its markup is therefore λ if the incumbent and the firm it displaced are of the same productivity type, $\lambda\varphi^h/\varphi^\ell$ if a high-type firm displaced a low-type firm, and $\lambda\varphi^\ell/\varphi^h$ in the reverse case. Let S_t denote the share of product lines operated by high-type firms. Because innovation is undirected, the displaced firm is of the high type with probability S_t, and the weight of the captured line is a draw from G. The expected profits of a newly captured product line are thus

$$\bar{\pi}_t^h = Y_t \left[1 - \frac{1}{\lambda}\left(S_t + (1 - S_t)\frac{\varphi^\ell}{\varphi^h}\right)\right] \quad \text{and} \quad \bar{\pi}_t^\ell = Y_t \left[1 - \frac{1}{\lambda}\left(S_t \frac{\varphi^h}{\varphi^\ell} + 1 - S_t\right)\right] \tag{6}$$

for a high-type and a low-type firm, respectively. For any S_t, a high-type firm earns more on a captured line than a low-type firm, and the difference increases in the productivity gap.

The share S_t is therefore a state variable of the firm's problem. Firms take it as given, as they do aggregate output, the wage, the interest rate, and the rate of creative destruction. It evolves according to

$$\dot{S}_t = S_t x_t^h + z_t p^h - S_t \tau_t, \tag{7}$$

where x_t^h denotes the expansion rate of high-type firms: these firms gain product lines through their own expansion and through the entry of new high-type firms, and they lose them to creative destruction. Supplemental Appendix D.4 derives eq. (7) from the distribution of productivity gaps across product lines, which also determines the

[12] As in Aghion et al. (2023), I assume that the step size of quality improvements exceeds the productivity differential, $\lambda > \varphi^h/\varphi^\ell$. This assumption ensures that the firm with the highest quality version in a product line is the incumbent producer. Relaxing this assumption would give room for a race for incumbency between low-productivity entrants and high-productivity incumbents, from which I abstract. The parameter assumption is fulfilled in all model estimations.

distribution of markups.

Firms choose their expansion rates to maximize the present value of profits net of R&D costs; Supplemental Appendix D.2 states the Hamilton-Jacobi-Bellman equations of the two types. Entry is free. When $z_t > 0$, the expected value of an entrant equals the cost of entry,

$$p^h E[V_t^h(1, \mu_i, \beta_i)] + (1 - p^h) E[V_t^\ell(1, \mu_i, \beta_i)] = \frac{1}{\psi_z} w_t, \tag{8}$$

where $E[V_t^d(1, \mu_i, \beta_i)]$ denotes the expected value of a firm of type $d \in \{h, \ell\}$ with one newly captured product. Lastly, the labor market clears: production labor and the research labor of incumbents and entrants add up to the aggregate labor endowment,

$$1 = \frac{Y_t}{w_t} \int_0^1 \mu_i^{-1} di + \int_0^1 \frac{x_{it}^\zeta}{\psi_x} di + \frac{z_t}{\psi_z}. \tag{9}$$

5.4 Balanced growth path

Along a balanced growth path, aggregate variables grow at a constant rate and the distribution of quality and productivity gaps across product lines is stationary, so that S, τ, and z are constant (Supplemental Appendix D.2 states the definition). The balanced growth path can be characterized in closed form.

Proposition 1. *Along a balanced growth path:*

1. *The value of a product line i for firm-productivity type $d \in \{h, \ell\}$ is given by*

$$V_i^d(1, \mu_i, \beta_i, S) = \frac{1}{\rho + \tau} \left[\beta_i Y_t \left(1 - \frac{1}{\mu_i} \right) + \frac{\zeta - 1}{\psi_x} (x^d)^\zeta w_t \right]. \tag{10}$$

   The value of a firm is the sum of the values of its product lines.

2. *Expansion R&D rates are firm productivity-type specific, i.e., $x_i \in \{x^h, x^\ell\}$, and equate the marginal cost of expansion R&D with the expected value of a newly captured product line,*

$$\frac{\zeta}{\psi_x} (x^d)^{\zeta - 1} w_t = \frac{1}{\rho + \tau} \left[\bar{\pi}_t^d + \frac{\zeta - 1}{\psi_x} (x^d)^\zeta w_t \right]. \tag{11}$$

   High-productivity firms expand faster than low-productivity ones, $x^h > x^\ell$.

3. *The share of product lines operated by high-productivity firms is* $S = zp^h/(\tau - x^h)$.

4. *The growth rate of aggregate variables is given by*

$$g = \frac{\dot{Q}_t}{Q_t} = \tau \times \ln(\lambda) = \Bigg(\underbrace{Sx^h + (1-S)x^\ell}_{\textit{Incumbent expansion R\&D}} + \underbrace{z}_{\textit{Entry}} \Bigg) \times \ln(\lambda). \tag{12}$$

5. *The expansion R&D rates* x^h *and* x^ℓ*, the entry rate* z*, the rate of creative destruction* τ*, the share* S*, the growth rate* g*, and the distribution of the number of products across firms do not depend on the distribution* G *of the product-line weights. The weights affect only the distribution of firm sales conditional on the number of products.*

Supplemental Appendices D.2, D.3, and D.4 contain the proofs. The value of a product line in eq. (10) consists of the profits for a given markup and the continuation value of expansion R&D, discounted at the sum of the discount rate and the rate of creative destruction, at which the line is lost. Productivity types make the incentive to expand type specific. The expected profits of a newly captured line are higher for a high-type firm, eq. (6), so that the optimality condition (11) holds at a higher expansion rate for the high type. The difference between x^h and x^ℓ is the channel through which the productivity gap shapes firm dynamics.

The expression for S follows from setting the law of motion (7) to zero: along the balanced growth path, the product lines that high-type firms lose to creative destruction, $S\tau$, equal those they gain through expansion and through entry, $Sx^h + zp^h$. Because high-type firms expand faster, the two types coexist only if entry is positive; without entry, high-type firms would eventually operate all product lines.

Long-run growth results from creative destruction: the growth rate in eq. (12) is the rate of creative destruction multiplied by the log step size of innovation. Since expansion rates differ between the types, growth depends on the allocation of product lines between them, S, in addition to the expansion and entry rates. Supplemental Appendix D.4 characterizes the aggregate labor income share, the misallocation measure $\mathcal{M}$, and the aggregate markup.

The last item of Proposition 1 states that the product-line weights are neutral for everything but the cross-section of firm sales. The reason is visible in eq. (10): the weight scales the profits of a product line but not the value of the expansion option attached to it, so the value of a product line is linear in its weight. Because innovation is undirected, the weight of a newly acquired product is a draw from G, and its expected value equals the value of a product with the mean weight of one. The opti-

mality condition for expansion R&D and the free-entry condition therefore coincide with those under equal weights, and so do all aggregate outcomes, since the weights are independent of markups across products. The same holds along a transition between balanced growth paths. This neutrality is what allows the quantitative analysis to discipline the *level* of size dispersion with the distribution of weights while leaving the mechanism of interest — the effect of the productivity gap on firm dynamics — untouched.

To find the balanced growth path, I jointly solve the optimality conditions of the two types, eq. (11), the free-entry condition, eq. (8), the labor market clearing condition, eq. (9), and the stationary distribution of productivity gaps across product lines (Supplemental Appendix D.4).

5.5 Firm dynamics

Conditional on its productivity type, a firm gains and loses products as in Klette and Kortum (2004): each of its products generates a new product at rate x^d and is lost at rate τ, and the firm exits when it loses its last product.[13] A firm of type $d \in \{h, \ell\}$ thus survives until age a_f with probability $\chi^d(a_f)$ and, conditional on survival, sells n products with probability $\left(1-\gamma^d(a_f)\right)\left(\gamma^d(a_f)\right)^{n-1}$, where

$$\gamma^d(a_f) = x^d \frac{1-e^{-(\tau-x^d)a_f}}{\tau - x^d e^{-(\tau-x^d)a_f}} \quad \text{and} \quad \chi^d(a_f) = 1 - \frac{\tau}{x^d}\gamma^d(a_f). \tag{13}$$

What productivity types add is that these dynamics differ across firms of the same cohort. Since $x^h > x^\ell$, high-type firms accumulate products faster and survive longer than low-type firms, so that the share of high-type firms among the surviving firms of a cohort,

$$s^h(a_f) = \frac{p^h \chi^h(a_f)}{p^h \chi^h(a_f) + (1-p^h)\chi^\ell(a_f)}, \tag{14}$$

increases with age.

The product-line weights map the number of products into sales, eq. (3). Because innovation and creative destruction are undirected, the weights of a surviving firm's products are independent draws from G with mean one. The average sales of surviving firms of type d at age a_f relative to the average sales of entrants therefore equal the

[13] The properties of firm growth and survival in Klette and Kortum (2004) therefore hold conditional on the firm type. In particular, conditional on the type, firm size and growth are unrelated, as in Lentz and Mortensen (2008).

average number of products,

$$\frac{E\left[\text{Size}_f|a_f,\varphi^d\right]}{E\left[\text{Size}_f|0,\varphi^d\right]} = \underbrace{E\left[n_f|a_f,\varphi^d\right]}_{\text{Avg. product count}} = \frac{1}{1-\gamma^d(a_f)}. \tag{15}$$

Pooled across the two types, the log of this ratio is the model counterpart of Figure 2, and it does not depend on the distribution of the weights. The dispersion of sales among firms of the same age, in contrast, reflects both margins of eq. (3): the dispersion of the number of products, which the productivity gap shapes through differential expansion, and the dispersion of the weights. Since the expected markup is the same on every product line of a firm, eq. (15) also gives the average employment of surviving firms relative to that of entrants. Supplemental Appendix D.5 derives the firm-age distribution and size dispersion conditional on age from these objects and the distribution of the weights.

6 Model application

This section confronts the model with the data and asks how far rising productivity dispersion goes in explaining the observed trends in firm dynamics. I first compare balanced growth paths. I calibrate the model along an initial balanced growth path to the ingredients of the firm-size decomposition in eq. (1) and to macroeconomic conditions in Sweden around the turn of the millennium. For a new balanced growth path, I raise the productivity gap to match the documented changes and test it against alternative shifters. I then solve for the transition between the two balanced growth paths and evaluate the welfare consequences of the rise.

6.1 Balanced growth paths

Initial balanced growth path The model has eight parameters: the expansion R&D efficiency ψ_x, the innovation-cost curvature ζ, the entry efficiency ψ_z, the step size of quality improvements λ, the productivity differential φ^h/φ^ℓ, the share of high-productivity firms among entrants p^h, the tail index α of the product-line weights, and the discount rate ρ. I estimate four of these and set the remaining four outside the model. Two of the latter are standard: the discount rate ρ is fixed at 0.05 and, following Acemoglu et al. (2018), the cost curvature ζ is set to two, in line with evidence from the microeconometric innovation literature (Blundell et al., 2002; Hall and Ziedonis, 2001).

Three of the estimated parameters — ψ_x, ψ_z, and λ — are pinned down jointly by

three targets: the firm size–age relationship, the share of young firms, and aggregate TFP growth. Each maps tightly to one target. The fourth, the tail index α, follows afterwards from the level of size dispersion conditional on age.

The size–age relationship identifies the expansion efficiency ψ_x. Successful expansion R&D adds product lines and thereby raises a firm's sales, so the speed at which firm size grows with age pins down how cheaply firms expand: a higher ψ_x lowers the cost of expansion R&D and steepens the size–age profile. I target the log of average size among firms aged six to ten relative to that of entrants of age zero, as displayed in Figure 2, a value of 0.501 log points at the start of the sample.

The entry efficiency ψ_z is identified by the share of young firms. Since ψ_z governs the labor cost of sustaining a given flow of entrants, it maps directly into the mass of recently born firms. I target the share of firms aged zero or one (Figure 3), which stood at 14.9% in 1997.

Finally, the step size of quality improvements λ is identified by aggregate TFP growth, on which it bears directly through the growth equation (12). I target a growth rate of 1.5%, the Swedish average over 1995–2005 (FRED).

The productivity gap φ^h/φ^ℓ and the share of high-type entrants p^h would naturally be identified from the level of size dispersion conditional on age. There are, however, good reasons not to load the level of dispersion on the productivity gap. Hottman et al. (2016) attribute 76% of the variance of log firm sales among U.S. consumer goods firms to firm appeal and 21% to product scope, while differences in average marginal cost contribute nothing; their two cost terms together account for only 3%.[14] The Swedish data point the same way: sales per product are considerably more dispersed across firms than product counts (Figure 5). Productivity heterogeneity thus likely accounts for only a small share of the level of size dispersion. The model confirms this. With productivity gaps but *equal* product-line weights, it has no dispersion in sales per product, and the gap cannot make up for it: for sufficiently large gaps, entry converges to zero and the balanced growth path ceases to exist long before empirical dispersion levels are reached.[15] In this respect the model is in the tradition of canonical models of firm dynamics built on creative destruction, which are well

[14] In the model in this paper, productivity differences also show up indirectly in product scope, as more productive firms add products faster. But already in a standard Klette and Kortum (2004) economy, ex-ante identical firms differ widely in scope, as products arrive and are lost at random, so productivity differences can account for at most a fraction of the scope share.

[15] This failure persists even when all model parameters are included in the estimation. Nor does adding further firm types help. Every firm enters with a single product and types differ in sales only through their number of products, so the dispersion of product counts within an age group is bounded by the count process of the most expansive type, whose expansion rate must remain below the rate of creative destruction for entry to stay positive; additional types only subdivide a range that two types already span.

known to struggle to produce realistic size dispersion (Luttmer, 2011; Berlingieri et al., 2024).

I therefore let the product-line weights carry the level of dispersion and set φ^h/φ^ℓ equal to unity in the initial balanced growth path, so that the rising gaps studied below are deviations from this homogeneous benchmark. When the two types are equally productive, p^h is immaterial. The level of size dispersion conditional on age then identifies the tail index α of the product-line weights. I assume that the distribution G of the weights is Pareto,[16] and a lower α spreads the sales of firms with the same number of products further apart. I target the standard deviation of log sales among firms aged six to ten, which stands at 1.80 in the first year in which this group is observed (Figure A-3). Because the weights leave the equilibrium and the three moments above unchanged (Proposition 1), ψ_x, ψ_z, and λ are estimated independently of α, which is then set to match the dispersion target.

The converse question is why the *rise* in dispersion should then be attributed to the productivity gap rather than to more dispersed weights. The two leave different footprints across firm ages. More dispersed weights raise size dispersion at every age, and one for one among entrants, who hold a single product. A wider gap is silent at entry and raises dispersion only as differences in product counts cumulate with tenure. In the data, dispersion among entrants and young firms is flat (Figure 4), and so is the dispersion of sales per product (Figure 5, panel b). I therefore hold the distribution of weights fixed throughout. Table 3 summarizes the targets and the resulting estimates.

The model matches all four targeted moments exactly.[17] One estimated parameter admits a direct interpretation: a successful innovation raises product quality by 10.6% ($\lambda = 1.106$). Because the initial balanced growth path features no productivity heterogeneity, every product earns the same markup, so the aggregate markup — cost- and sales-weighted measures coincide — equals 1.106 as well. Before turning to the main exercise, I use a set of comparative statics in the productivity gap to illustrate the model's mechanics.

Figures 6 and 7 show size dispersion conditional on age and the firm-age distribution for a range of productivity gaps, starting from the initial balanced growth path

[16] Since the estimated tail index is below one, I bound the support of the weights from above, at 10^{10} times its lower bound, so that their mean is finite, as the model requires. The bound is large enough not to affect the size of any firm; the results are unchanged at 10^{12}. Rescaling the weights to a mean of one leaves the dispersion of log sales unchanged.

[17] The three parameters minimize the sum of squared percentage deviations from their targets. A global step evaluates this loss at 12,000 parameter vectors from a Sobol sequence, and a local step refines the 50 best candidates; every local search converges to the same parameter vector. Given the three parameters, the tail index α solves the fourth moment condition.

Table 3: Initial balanced growth path. Moments and parameters

	Data	Model
Moments		
Size-age: 6–10 vs. entrants in logs	0.501	0.501
Young-firm share in %	14.9	14.9
TFP growth g in %	1.5	1.5
Size dispersion: SD of log sales, ages 6–10	1.80	1.80
Parameters		
ψ_x *Expansion R&D efficiency*		0.311
ψ_z *Entry R&D efficiency*		1.472
λ *Step size of quality improvements*		1.106
α *Tail index of product-line weights*		0.630
Set exogenously		
ρ *Discount rate*		0.05
ζ *R&D cost curvature*		2
φ^h/φ^ℓ *Productivity differential*		1

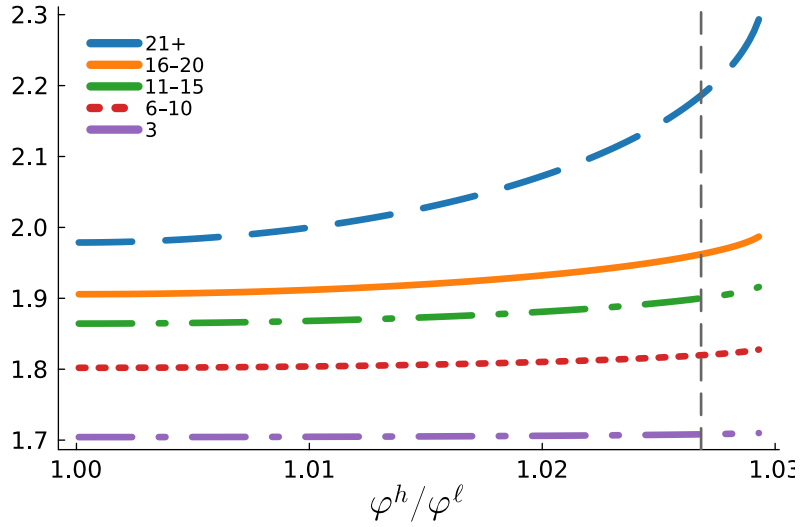
Notes: the size-age relationship measures the log of average size of surviving firms aged six to ten relative to that of entrants (age zero) (Figure 2). The young-firm share measures the share of firms aged zero or one. Size dispersion is the standard deviation of log sales among firms aged six to ten in the first year in which this group is observed (Figure A-3). The targets are computed using Swedish registry data, except for aggregate productivity (TFP) growth (FRED).

($\varphi^h/\varphi^\ell = 1$). Once the gap exceeds unity, the share of high-type entrants matters, and I set it exogenously to $p^h = 0.5$, so that entrants are equally likely to be born of either type. Figure 6 measures dispersion by the standard deviation of log sales.[18] Raising the gap steepens the age profile of dispersion: dispersion rises markedly among older firms and stays flat among young firms. The mechanism is differential expansion. A larger productivity gap raises the value of a high-type's product, and with it the high type's incentive to expand, so more productive firms expand faster than less productive ones, and the resulting size differences cumulate with tenure. This age-dependent rise mirrors the empirical pattern (Figure A-3).

Figure 7 turns to the firm-age distribution over the same range of gaps. At the initial balanced growth path, the share of young firms in the left panel matches its target of 14.9%. The share of firms at least ten years old in the right panel is untargeted: the model (50%) slightly overstates the data (45%; Figure 3). As the productivity gap rises, the share of young firms falls and the share of old firms rises. The mechanism

[18] I use the standard deviation rather than the IQR because the variance is the object of the decomposition in eq. (1).

Figure 6: Firm-size dispersion conditional on age (model)



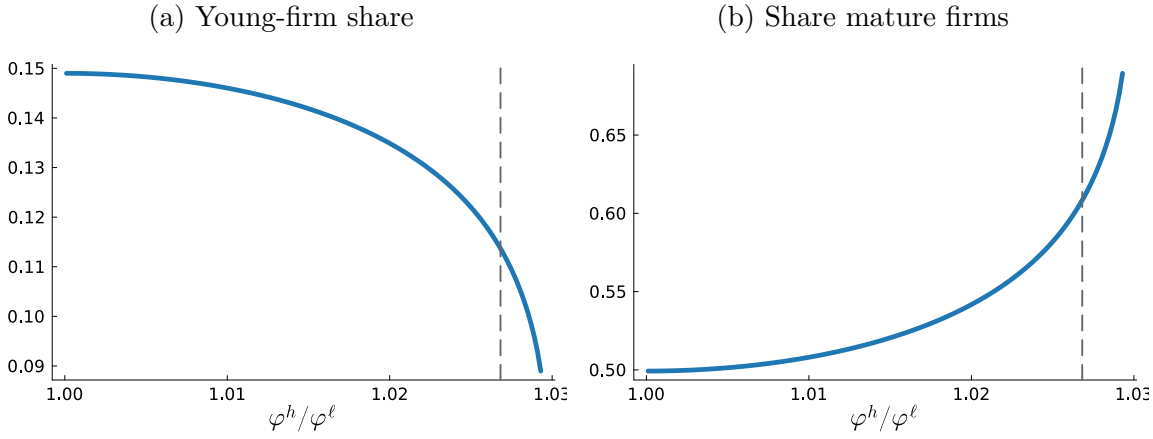
Notes: The figure shows the standard deviation of log firm size (sales) by firm age indicated in the legend. The figure starts from the initial balanced growth path ($\varphi^h/\varphi^\ell = 1$) and then continuously raises the gap. The vertical dashed line marks the estimated new-balanced-growth-path gap ($\varphi^h/\varphi^\ell = 1.027$).

is again differential expansion, now working through firm survival. A firm loses each of its products to creative destruction at rate τ and replaces products by expanding at rate x, so the difference $\tau - x$ governs how quickly it is whittled down to exit. As the productivity gap widens, x^h rises toward τ: high-type firms come to replace destroyed products almost as fast as they lose them and survive for longer. Because a fixed share of entrants are high type, these long-lived firms accumulate in the upper tail of the age distribution, raising the share of old firms. At the same time, the lower tail of the age distribution thins: with high-type firms surviving for longer, firms live longer on average, and recent cohorts make up a smaller share of the population. Over the displayed range of gaps ($\approx 2.9\%$), the share of young firms falls by six percentage points and the share of old firms rises by 19.

The mean size–age relationship, by contrast, is largely insensitive to the gap. Faster expansion by high-productivity firms is offset by slower expansion among low-productivity firms, so the net effect on the average size–age profile is small. Across the same range of gaps, the size difference between firms aged six to ten and entrants — targeted at 0.501 log points in the initial balanced growth path — rises only to 0.555, consistent with the essentially flat mean size–age profile in the data.

New balanced growth path I now raise the productivity gap to match the two documented changes in firm dynamics, holding all other parameters at their initial values. The first is the rise in size dispersion among older firms: the standard devia-

Figure 7: Firm-age distribution (model)



Notes: The left panel shows the young-firm share, defined as the share of firms aged zero or one. The right panel shows the share of firms that are at least ten years old. Both panels start from the initial balanced growth path ($\varphi^h/\varphi^\ell = 1$) and then continuously raise the gap. The vertical dashed line marks the estimated new-balanced-growth-path gap ($\varphi^h/\varphi^\ell = 1.027$).

tion of log sales rose by roughly 0.2 log points for the older age groups (Figure A-3), which I target for firms aged 21 and above. The second is the decline in the young-firm share, from 14.9% in 1997 to 10.4% in 2017 (Figure 3). The test is demanding: one parameter must match two margins of the data at once. The share of high-type entrants stays at $p^h = 0.5$, as in the comparative statics above. Supplemental Appendix E repeats the estimation for values of p^h between 0.4 and 0.8 — the range in which the targeted rise in old-age dispersion is attainable, since that rise requires both types to be present among old firms — and shows that only the inferred size of the rise in the productivity gap depends on this value; the results that follow do not.

The first productivity-gap column of Table 4 reports the results. The productivity gap rises from unity in the initial balanced growth path to 1.027. Modest as this increase appears, it opens a sizable wedge between the expansion rates of the two types ($x^h = 0.129$ and $x^\ell = 0.082$, against 0.106 for both types initially), and the size differences this wedge generates over the life cycle are enough to reproduce the targeted rise in dispersion among old firms.[19] The standard deviation of log size for firms aged 21 and above rises from 1.98 to 2.19 — an increase of 0.21 log points, against the targeted 0.2 — and the share of young firms falls to 11.4%, one percentage point short of its target of 10.4%. The share of firms at least ten years old, which is not targeted, rises by 11 percentage points, close to the 12-point increase in the data

[19] In a similar vein, Aghion et al. (2023) find that a small, 4% increase in the productivity gap more than accounts for the long-run fall in U.S. productivity growth.

(Figure 3). A single parameter thus accounts for the rise in within-age dispersion and for most of the shift in the age distribution — a nontrivial result, since the productivity gap is overidentified by the two targets.[20] The second column replaces the dispersion target by the rise in the right tail, the gap between the 99th and the 25th percentile of log sales among firms aged 21 and above, which widened by 0.48 log points in the data (Figure A-9); the inferred gap, 1.028, and its implications are nearly the same. The tail target matches the decline in the young-firm share even more closely: the share falls to 10.6%, 0.2 percentage points short of its target.

As a further untargeted implication, the rise in the productivity gap slows long-run growth: aggregate TFP growth falls by 0.10 percentage points relative to its baseline of 1.5%. By eq. (12), growth is proportional to the rate of creative destruction, which falls from 0.149 to 0.139, and the decline can be traced to its sources: expansion R&D by the two types, the allocation of product lines between them, and the flow rate of entry. Expansion R&D moves in opposite directions for the two types: high-type firms expand faster, and as product lines are reallocated toward them their expansion carries an ever larger weight in creative destruction, while low-type firms expand more slowly. Taken together, faster high-type expansion contributes +0.16 percentage points to growth, the reallocation of lines toward the faster-expanding type +0.08, and slower low-type expansion −0.08, so that creative destruction by incumbents rises on net, from 0.106 to 0.122.[21] The source of the slowdown is entry: the flow rate of entry z falls from 0.043 to 0.017 and contributes −0.26 percentage points, which outweighs the net incumbent gain. Entry declines for two reasons. Once high-type firms hold more than eight in ten product lines, a high-type entrant rarely captures a line from a less productive incumbent, where its advantage pays off, while a low-type entrant mostly faces a more productive incumbent and earns a compressed markup. Free entry then requires that product lines last longer, so the rate of creative destruction falls. In addition, incumbents account for a larger part of creative destruction as product lines shift toward the faster-expanding type, which crowds out entrants. The rise in size dispersion conditional on age is therefore consistent with a mild decline in long-run productivity growth — a connection whose welfare implications I take up in the transition analysis.

[20] The estimation attributes all of the shift in the age distribution to the rising productivity gap. A natural alternative would instead attribute the decline in entry, and the resulting aging of the firm population, to slower labor force growth. This confound is absent here: labor force growth in Sweden did not decline over the sample period (Engbom, 2023).

[21] The split evaluates eq. (12) at the averages of S, x^h, and x^ℓ across the two balanced growth paths, so that the contributions of entry, high-type expansion, low-type expansion, and the reallocation of product lines between the types sum exactly to the change in growth.

Table 4: New balanced growth path: the productivity gap versus alternative shifters

	Data	Prod. gap $\varphi^h/\varphi^\ell\uparrow$		R&D-cost gap	Cheaper exp.	Entry barr.
		SD target	Tail target	$\psi_x^h/\psi_x^\ell\uparrow$	$\psi_x\uparrow$	$\psi_z\downarrow$
Targeted						
$\Delta\sigma$ (ln Size|Age=21+)	0.20	0.21	(0.24)	0.21	0.21	0.19
Δ (P99 − P25) (ln Size|Age=21+)	0.48	(0.41)	0.48	(0.41)	(0.30)	(0.28)
Young-firm share in %	10.4	11.4	10.6	13.0	11.5	9.2
Untargeted						
$\Delta\sigma$ (ln Size|Age=3)	≈ 0	0.00	0.01	0.01	0.05	0.02
Size-age: 6–10 vs. entrants	0.50	0.54	0.54	0.55	0.73	0.63
Δ TFP growth in pp	−0.43	−0.10	−0.11	+0.04	+0.27	−0.12
Implied parameter		1.027	1.028	1.432	0.467	1.250

Notes: Each column re-estimates the parameter in its header to two targets — the rise in size dispersion among firms aged 21 and above and the young-firm share — with all other parameters at their values in the initial balanced growth path, where the young-firm share is 14.9%, size-age 0.50, $\psi_x = 0.311$, and $\psi_z = 1.472$. The dispersion target is the standard deviation of log sales; in the tail-target column it is the gap between the 99th and the 25th percentile of log sales, computed on unwinsorized sales over 2014–2017 (Figure A-9). Entries in parentheses are untargeted. The R&D-cost gap is a geometric mean-preserving spread of the expansion efficiency between the two types, at a productivity gap of unity. Size-age is the average size of firms aged six to ten relative to entrants, in logs; the data entry is the value at the start of the sample, and the profile is essentially flat in the data (Section 4). The change in TFP growth, in percentage points, compares the two balanced growth paths in the model and the averages over 1995–2005 and 2010–2015 in the data (FRED).

Distinguishing the productivity gap from alternative shifters A rising productivity gap *can* reproduce the trends, but this does not establish that it is the only force that could. Within the model, three alternatives are natural candidates. Two are common to all firms: a rise in expansion efficiency ψ_x — a cheaper technology for adding product lines — and a fall in entry efficiency ψ_z — the entry-barriers mechanism emphasized in the declining-business-dynamism literature. The third, like the productivity gap, separates the two types: a widening gap in expansion R&D efficiency, in the spirit of the innovative-capacity heterogeneity of Acemoglu et al. (2018). I let the efficiency differ by type, so that a type-θ firm incurs the cost $(x)^\zeta/\psi_x^\theta$ of expanding at rate x, and widen the ratio ψ_x^h/ψ_x^ℓ as a geometric mean-preserving spread around the estimated ψ_x: the high type's R&D becomes cheaper and the low type's dearer, holding the geometric mean of the two efficiencies fixed.[22] I subject each alternative to the same test as the productivity gap. Holding all other parameters at their initial values — and, for the three alternatives, the productivity gap at unity — I re-estimate the single parameter to the same two targets, then ask what it implies for the margins those targets leave free but that the data discipline: dispersion among *young* firms and the mean size–age profile (Section 4), both essentially flat in the data, and the change in aggregate productivity growth, which slowed over the period.

[22]The type-specific efficiency enters the model only through the expansion R&D cost; the first-order conditions of Supplemental Appendix D.2 hold with ψ_x replaced by ψ_x^θ.

Table 4 reports the results. With one parameter and two targets, no shifter matches both targets exactly. The productivity gap and cheaper expansion come closest, reproducing the rise in old-age dispersion with the young-firm share about one percentage point above its target. Entry barriers overshoot on the other side: by the time dispersion among old firms has risen by 0.19 log points, the young-firm share has already fallen to 9.2%. The R&D-cost gap fails on the young-firm share: it reproduces the rise in old-age dispersion, but this share falls only to 13.0%, less than half of the way to its target; pushing the young-firm share further requires a gap under which old-age dispersion overshoots the target substantially.

The two common shifters diverge from the productivity gap on the discriminating size margins. The two type-specific shifters hold young-firm dispersion flat (a rise of 0.00 and 0.01 log points at age three, against 0.05 under cheaper expansion and 0.02 under entry barriers) and the mean size–age profile close to flat (firms aged six to ten are 0.54 and 0.55 log points above entrants, against 0.73 and 0.63, relative to 0.50 in the data at the start of the sample). The reason is mechanical: both common shifters raise dispersion by speeding expansion for *all* firms, which widens the product-count distribution at every age and steepens the size–age profile; a gap between the types instead raises dispersion only through the divergence between them, which accrues with tenure and so concentrates among older firms while leaving the average profile flat.[23] The R&D-cost gap reproduces this age signature for a simple reason: in the model, every moment of firm size and age depends on the parameters only through the rate of creative destruction and the two types' expansion rates, and both gaps work through the same wedge between the latter. The size data therefore identify a widening gap in expansion rates between persistent firm types, not by themselves its source.

What separates the two sources lies outside the size data. The first is the entry margin, on which the R&D-cost gap already fell short above. The second is aggregate growth. Under the productivity gap, growth slows by 0.10 percentage points; under the R&D-cost gap it *rises* by 0.04 percentage points. The difference is in the nature of the high type's advantage. A productivity advantage pays off when a high-type firm captures a product line from a low-type firm; as low-type firms disappear, this premium erodes, so the high type's expansion rises by less than its advantage would suggest, and creative destruction by incumbents rises too little to offset the fall in entry — growth falls. An R&D cost advantage does not depend on whom a line is

[23] The right tail points the same way. Although the two common shifters are fitted to the rise in the standard deviation of log sales among firms aged 21 and above, they miss the rise in its right tail: the gap between the 99th and the 25th percentile widens by 0.30 log points under cheaper expansion and 0.28 under entry barriers, against 0.41 under both type-specific shifters and 0.48 in the data (Figure A-9).

captured from. R&D is on average as costly as before, but an ever larger share of product lines is held by the type that expands more cheaply, so creative destruction rises even as entry declines. Since productivity growth slowed over the period in which dispersion rose — from 1.5% over 1995–2005 to 1.07% over 2010–2015 — the direction of growth favors the productivity gap. Disciplined by the data on firm size and age alone, the productivity gap thus accounts for about a quarter of the observed long-run decline in growth; the remainder is left to forces these data do not identify. I further provide empirical evidence for the suggested mechanism. Section 7 shows that revenue and physical productivity dispersion did widen in the Swedish data. Among the candidate shifters considered, then, the two type-specific gaps reproduce the age signature of the data, and the productivity gap alone also matches the decline in the young-firm share, the direction of growth, and the observed rise in productivity dispersion.

6.2 Transition analysis

The balanced-growth-path analysis compared two long-run equilibria. I now solve for the full transition path between them, both to trace how the economy adjusts over time and to evaluate the welfare consequences of the rising productivity gap. Starting from the initial balanced growth path, I raise the productivity gap once and for all to its new-balanced-growth-path value as an unanticipated, permanent shock in period zero, holding all other parameters fixed. I model the increase in the gap as a (geometric) mean-preserving spread: as φ^h/φ^ℓ widens, the high type becomes more productive and the low type less productive, holding the geometric mean of the two types' productivities fixed. The wider gap is therefore pure productivity dispersion and does not raise average productivity; the *level* of aggregate productivity then moves only through the reallocation of market share toward the more productive type.[24]

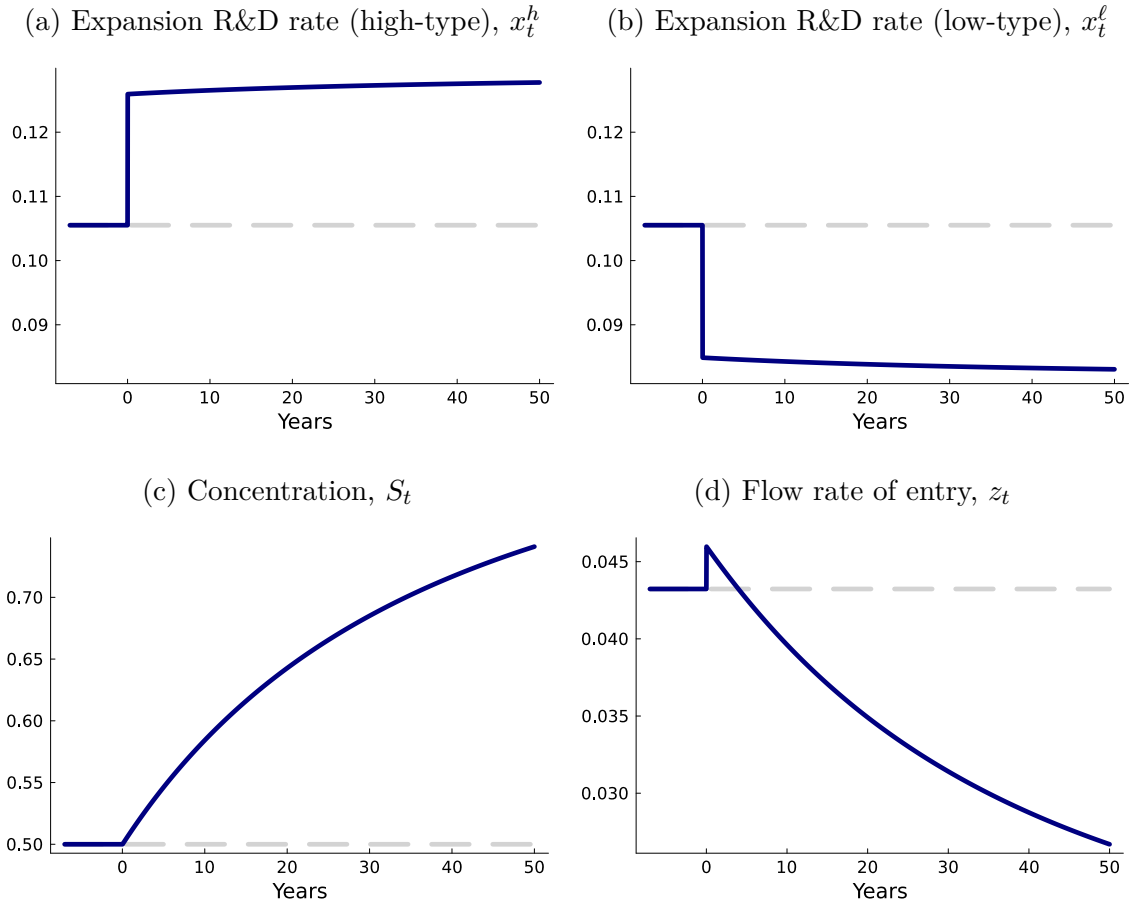
The sign of the welfare effect is unclear ex ante: slower long-run growth is costly to the representative consumer, whereas the reallocation of market share toward high-productivity firms raises the *level* of aggregate productivity. Which force dominates can only be settled by solving the transition.

I solve for the transition path as a fixed point in the sequences of wage growth, interest rates, and the market share of high-productivity firms, S_t. Given a guess, I solve firms' value and policy functions backward from the new balanced growth path, simulate the distribution of productivity gaps forward from the initial one, and

[24] All equilibrium objects depend on the two types' productivities only through their ratio φ^h/φ^ℓ; the mean-preserving-spread assumption therefore leaves the model solution unchanged and bears only on the level of aggregate productivity Φ, which depends on the productivities themselves.

back out the implied sequences from market clearing and the optimality conditions. Supplemental Appendix F describes the algorithm in detail.

Figure 8: Transitional dynamics for equilibrium outcomes

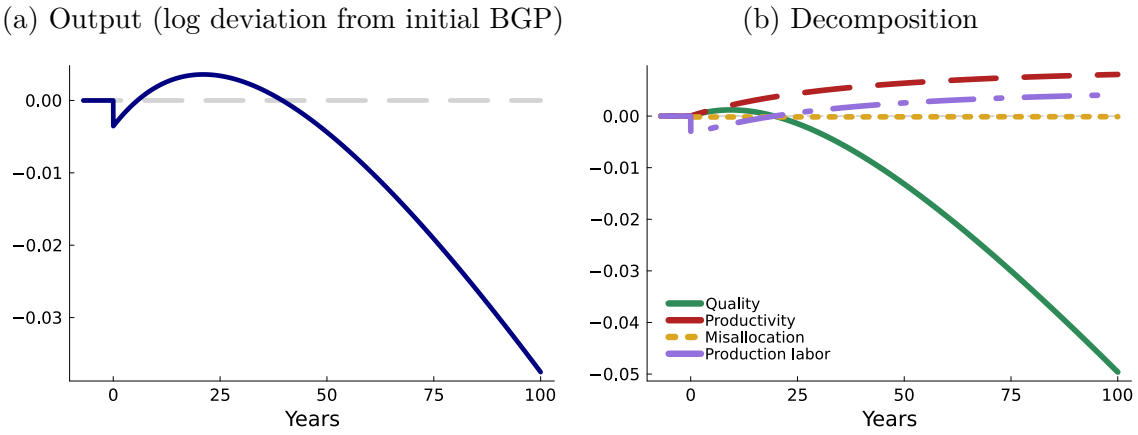


Notes: the figure shows the response in equilibrium outcomes during the transition toward the new balanced growth path. The productivity gap φ^h/φ^ℓ is raised from 1 to 1.027 (Table 4) as a one-time shock in period zero; all other parameters are held fixed. The dashed line indicates the initial balanced growth path.

Figure 8 shows the response of the main equilibrium outcomes over the transition. The four panels are the ingredients of the growth rate in eq. (12): x_t^h, x_t^ℓ, and S_t determine creative destruction by incumbents, and z_t the entry component. Because a larger productivity gap raises the value of a high-type firm's products, high-productivity firms expand more aggressively: x^h jumps from 0.106 to 0.126 on impact and edges up further to 0.129 over the transition. Low-productivity firms move in the opposite direction. Facing more productive competitors, their expansion rate x^ℓ falls on impact from the same value of 0.106 to 0.085; as these competitors also become more numerous, it declines further to 0.082. Most of the adjustment in the choice variables

occurs on impact; the remainder follows the gradual rise of S_t, with x^h reaching 0.127 after 25 years and 0.128 after a century. The widening gap in expansion rates reallocates market share toward the high type: the share of product lines operated by high-productivity firms, S_t, rises gradually from 0.5. The bottom-right panel shows the flow rate of entry z_t. Entry rises slightly on impact, from 0.043 to 0.046, as the higher gap raises the value of entering as a high-productivity firm. The increase is short-lived: as product lines shift toward high-type firms, the expected profits of an entrant's product fall and incumbents account for a larger part of creative destruction. The entry rate therefore declines steadily.

Figure 9: Transitional dynamics for aggregate output



Notes: Panel (a) shows the log deviation of aggregate output along the transition from the counterfactual path on which the economy remains on the initial balanced growth path. Panel (b) decomposes this deviation into the contributions of aggregate quality Q, productivity Φ, misallocation $\mathcal{M}$, and production labor L_P (eq. 4), each in log points relative to the initial balanced growth path. The productivity gap is raised from 1 to 1.027 in period zero.

Figure 9 turns to aggregate output. Panel (a) plots the log deviation of output along the transition from the counterfactual on which the economy would have remained on the initial balanced growth path, $\ln(Y_t/Y_t^{\text{init}})$ where Y_t^{init} denotes output on that path; removing the common trend isolates the effect of the shock. Output dips slightly on impact and a modest, temporary boom follows: the boom peaks at about 0.004 log points (0.4%) around year twenty before declining, crossing back below the initial-balanced-growth-path trend after roughly forty years and remaining below thereafter. In the long run, the deviation declines linearly, at the difference between the two long-run growth rates, $g^{\text{new}} - g^{\text{initial}} < 0$. Panel (b) decomposes this deviation using eq. (4). Since $Y = Q\Phi\mathcal{M}L_P$, in logs the deviation of output is the sum of the deviations of aggregate quality Q, productivity Φ, misallocation $\mathcal{M}$, and production labor L_P from their initial-balanced-growth-path values.

The decomposition reveals two offsetting forces. The level gain in aggregate output comes from aggregate productivity Φ, which rises steadily as an increasing share of

product lines is operated by the more productive type. This reallocation lifts aggregate output 0.008 log points (0.8%) above its initial balanced growth path value by year one hundred. Working in the opposite direction is aggregate quality Q. Its contribution is in fact slightly positive in the early years of the transition: on impact, entry z rises, while the faster expansion of high-type firms is offset by the slower expansion of low-type firms, so the aggregate rate of creative destruction — and with it quality growth — exceeds its initial-balanced-growth-path rate during the first nine years. This reverses as the transition proceeds: entry declines, outweighing the higher expansion intensity of the now-dominant high-productivity firms, so creative destruction falls below its initial level and quality grows more slowly thereafter. The resulting quality shortfall is negligible at first but cumulates over time, reaching -0.050 log points (-5.0%) by year one hundred, and is the force that eventually pulls output below trend. Misallocation $\mathcal{M}$ contributes negligibly. Production labor L_P falls on impact — labor shifts out of production and into entry and expansion R&D — before recovering and turning mildly positive; this fall in production labor is what makes output dip slightly on impact. In sum, the rising productivity gap produces a modest short-run boom through the reallocation of market share toward more productive firms, but a long-run decline through permanently slower growth from reduced creative destruction.[25]

To weigh the short-run boom against the long-run decline, I compute the permanent percentage change in consumption along the initial balanced growth path that makes the representative consumer indifferent to the consumption stream realized over the transition toward the new balanced growth path. Accounting for the transition, welfare *falls* by only 0.042%: the level gain from reallocating market shares toward more productive firms offsets most — but not quite all — of the discounted cost of slower long-run growth, leaving a small net loss.[26] Ignoring the transition substantially overstates this loss. Comparing the two balanced growth paths directly,

[25] The same anatomy, seen from asset prices, is what Paron (2026) documents for the United States: most of the rise in aggregate valuation ratios reflects a reallocation of sales toward high-valuation firms, and in his Schumpeterian model a decline in the productivity of innovation concentrates production in the most efficient incumbents, so that stock-market wealth booms while entry and growth fall and the present value of consumption stagnates.

[26] Aghion et al. (2023) find a larger welfare loss, 3.3%, from a rise in the productivity gap identified by the rise in top-firm concentration. The difference lies in the margin of adjustment rather than in the target. In both models, the rise of the high type compresses markups and lowers the profit of a product line, which in equilibrium is offset by lines lasting longer, that is, by less creative destruction. In their model, firms neither enter nor exit, so creative destruction consists of incumbent innovation alone: both types innovate less, and growth falls by 0.6 percentage points for a 4% rise in the gap. Here, free entry ties the value of a line to the cost of entry and thereby pins the average of the two types' expansion rates, so the adjustment runs through entry: the flow of entrants falls, lines last longer, and incumbents' average expansion rate is unchanged, leaving a growth loss of 0.1 percentage points.

the consumption-equivalent change ξ solves $\ln(1+\xi) = (g^{\text{new}} - g^{\text{initial}})/\rho$; with g falling by 0.10 percentage points, this across-balanced-growth-path comparison implies a welfare loss of 1.92% ($\xi = -0.0192$). The comparison misses two features of the transition. First, the slowdown does not arrive at once: creative destruction initially exceeds its previous rate and declines only as product lines shift toward high-type firms. Second, the reallocation of market share raises the level of aggregate productivity. The choice of target matters little for this conclusion: re-estimating the gap to the rise in the extreme right tail of the size distribution among old firms instead (the tail-target column of Table 4) gives a slightly larger gap, a growth loss of 0.11 percentage points, and a welfare cost of 0.046% with the transition (2.26% across balanced growth paths).

7 The mechanism in the data

The empirical part of the paper documented the consequences of the mechanism — the margins of firm dynamics along which size dispersion rose — and the model showed that a widening productivity gap between firms reproduces them. This section turns to the cause and asks whether the exogenous shifter of the model, dispersion in productivity across firms, has in fact widened in the same data.

The model's productivity gap is a gap in physical productivity, which balance-sheet data do not observe. What they do observe is revenue productivity: the value of output relative to inputs. I measure it as a cost-share Solow residual on gross output,

$$\ln \text{TFPR}_{ft} = \ln \text{PV}_{ft} - s_{jt}^{L} \ln \text{WB}_{ft} - s_{jt}^{K} \ln K_{ft} - s_{jt}^{M} \ln M_{ft}, \tag{16}$$

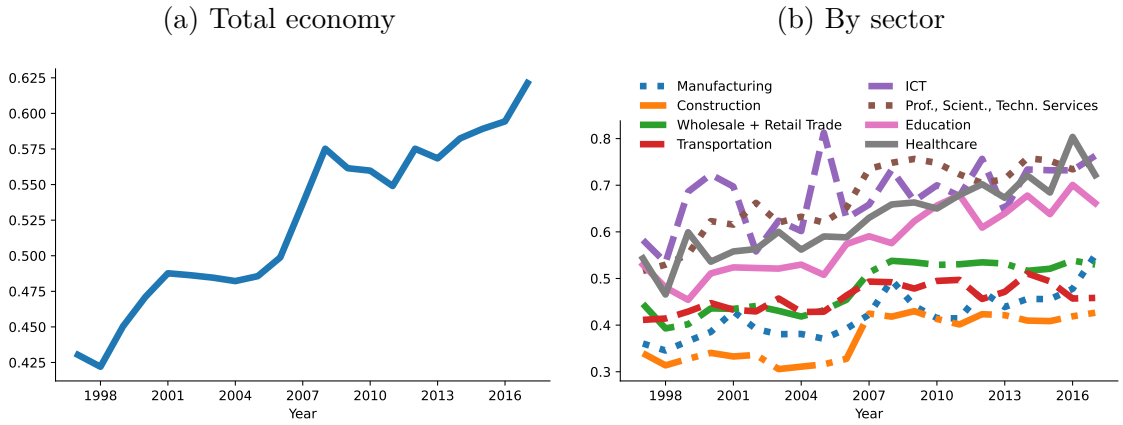
where PV_{ft} is the production value of firm f in year t, WB_{ft} its wage bill, K_{ft} its capital stock, and M_{ft} its intermediate inputs. The cost shares s_{jt}^{L}, s_{jt}^{K}, and s_{jt}^{M} are computed for each two-digit industry j and year t on the universe of firms, as the industry's total wage bill, capital cost, and intermediate inputs relative to their sum, with the capital cost set to ten percent of the capital stock. The sample is that of Section 3, restricted to firm-years with positive values of all four variables.

Revenue productivity is an imperfect measure of firm productivity. If firm f produces output $y_{ft} = \varphi_{ft} F(\cdot)$ with physical productivity φ_{ft} and a technology F common to the firms of its industry, its revenue per unit of inputs is $p_{ft} y_{ft} / F(\cdot) = p_{ft} \varphi_{ft}$: revenue productivity is the product of the firm's price and its physical productivity, and eq. (16) measures the log of this product. Its dispersion across firms therefore widens when physical productivity spreads out, but also when prices do. The advantage

of the measure is its coverage: it requires only balance-sheet data, so the evidence below covers firms of all sectors, sizes, and ages, as do the size trends of Section 4. Firm-level prices are observed only for manufacturing firms in the survey of industrial production; for this subset, I remove the price component below.

Dispersion is measured within industries. For each two-digit industry and year, I compute the standard deviation of log revenue productivity across firms in cells with at least ten firms, and, for each year, average the cell-level standard deviations across all industries or the industries within a sector, weighting each industry by its total costs.

Figure 10: Dispersion of revenue productivity within industries



Notes: The figure shows the standard deviation of log revenue productivity, eq. (16), across firms within two-digit industries. The standard deviation is computed for each two-digit industry and year (cells with at least ten firms) and averaged across industries with the industries' total costs as weights. Panel (a) averages over all industries; panel (b) averages over the two-digit industries of each sector. Cost shares are computed by two-digit industry and year on the universe of firms. Supplemental Appendix C reports the interquartile range and the series for firms aged zero to five.

Figure 10 reports the results. Panel (a) shows that revenue productivity dispersion within industries widened over the sample period: the standard deviation of log revenue productivity rises from 0.43 log points in 1997 to 0.50 in 2006 and 0.62 in 2017. The interquartile range widens as well, from 0.38 to 0.46 log points (Supplemental Appendix C), so the rise is not confined to the tails of the distribution. Panel (b) shows the same series by sector. Dispersion rises in every sector, by between 0.05 log points in Transportation and 0.24 in Professional, Scientific, and Technical Services, with Manufacturing, ICT, and Healthcare each close to 0.2. As with size dispersion, the widening of productivity dispersion is thus common to all sectors.

The cleanest cell for the mechanism is that of young firms. In the model, a firm's productivity type is assigned at entry, and the type composition of a cohort is subsequently shaped by the selection of survivors. Among firms aged five or younger,

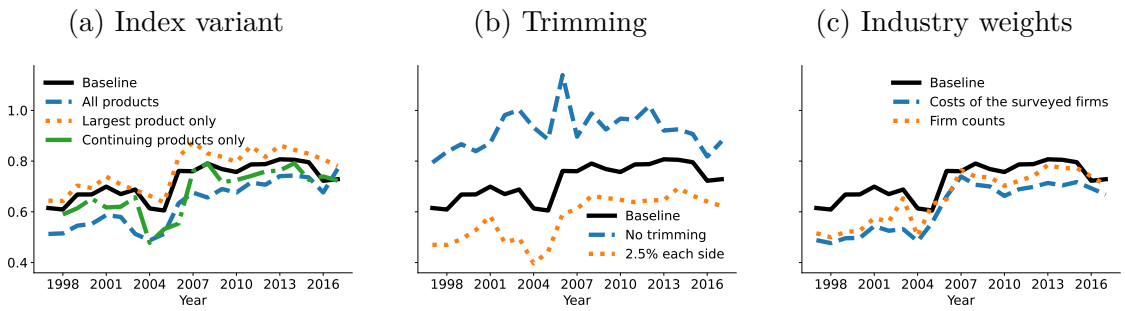
selection has had little time to operate, so the dispersion of their productivity comes closest to the dispersion with which firms enter. Figure A-10 (Supplemental Appendix) shows that the standard deviation of log revenue productivity among firms aged zero to five rose from 0.48 log points in 1997 to 0.68 in 2017, as much as in the total population. Measured by revenue productivity, the exogenous shifter of the model thus moved in the assumed direction, and it did so among young cohorts, not only among the old survivors. Read together with Figure 4, this is the signature of the mechanism: firms of the same age became more dispersed in productivity already when young, whereas the size differences this generates appear only with tenure, as the differential expansion of the two types cumulates.

For the manufacturing firms in the survey of industrial production, the price component can be removed. The survey records, for each firm, product code, and year, the value and the quantity delivered, so that a unit value is observed for every product a firm sells. I construct firm-level prices following Aghion et al. (2025), whose price index is that of De Ridder et al. (2026): each log unit value is expressed relative to the value-weighted mean log unit value of the firms selling the same product in the same year, and a firm's log price $\ln p_{ft}$ is the value-weighted average of these relative prices across its products. Physical productivity then follows as $\ln \mathrm{TFPQ}_{ft} = \ln \mathrm{TFPR}_{ft} - \ln p_{ft}$. A relative price is informative only for products with at least two sellers, so the baseline construction excludes the products of which a firm is the sole seller, trims the one percent most extreme relative prices on each side, and requires that the products with observed quantities cover at least half of a firm's production value; observations with imputed quantities are excluded. The sample consists of those firms of the product-count evidence of Section 4.4, surveyed manufacturing firms with at least 20 employees, for which a price and revenue productivity can be constructed. Dispersion is measured as for revenue productivity.

Because several of these choices are matters of judgment, Figure 11 reports the baseline together with one departure from it per panel: the treatment of single-seller products and of a firm's product portfolio (panel a), the trimming of relative prices (panel b), and the weights with which industries are averaged (panel c); Supplemental Appendix C adds further departures. In the baseline, the standard deviation of log physical productivity rises from 0.62 log points in 1997 to 0.73 in 2017; across all constructions the increase over the same period lies between 0.08 and 0.27 log points. Physical productivity dispersion among surveyed manufacturers thus widened over the period; by how much depends on the construction.

I read this evidence, like the product-scope and R&D evidence of Section 4.4, for its direction rather than for its magnitude. Product scope, R&D, and physical productivity are observed only for surveyed manufacturers or for larger R&D performers, and revenue productivity, available across all sectors and firm sizes, conflates prices

Figure 11: Dispersion of physical productivity among surveyed manufacturers



Notes: The figure shows the standard deviation of log physical productivity, $\ln \mathrm{TFPQ}_{ft} = \ln \mathrm{TFPR}_{ft} - \ln p_{ft}$, across surveyed manufacturing firms within two-digit industries, computed for each industry and year (cells with at least ten firms) and averaged across industries with the industries' total costs as weights. The baseline, repeated in each panel, uses the products with at least two sellers, trims relative prices at the first and 99th percentiles, and requires the products with observed quantities to cover at least half of a firm's production value. Panel (a) instead includes single-seller products at a log relative price of zero, uses only a firm's largest product, or only the products the firm also reported in the previous year. Panel (b) varies the trimming. Panel (c) averages industries with the costs of the surveyed firms or with firm counts. Supplemental Appendix C varies the coverage requirement, the cell threshold, and the sample.

with physical productivity. None of this evidence therefore enters the calibration, whose targets are moments of firm size, directly observed for the universe of firms. The survey and productivity evidence instead serves as an independent check on the mechanism. What that check requires is that firms of the same age came to differ more in their innovation effort and product scope, and that productivity dispersion widened over the period in which their sizes diverged. This is what the data suggest.

8 Conclusion

This paper documents a sustained rise in firm-size dispersion in Sweden from 1997 to 2017 and decomposes it, using the law of total variance, into three margins of firm dynamics: differences in mean size across firm ages, the firm-age distribution, and size dispersion conditional on age. Over this period the age distribution shifted toward older firms, mean size by age changed little, and size dispersion conditional on age rose, concentrated among older firms. The decomposition — model-free and exact in the data — attributes the overwhelming share of the rise in dispersion to rising size dispersion conditional on age, while the aging of the firm population contributes modestly and changes in mean size by age contribute negligibly.

The changes in these three margins provide disciplining moments that any candidate mechanism of rising firm-size dispersion should reproduce. The concentration of the rise in dispersion among older firms is an additional discriminating fact: it points

to forces whose effect accumulates with firm age, such as widening productivity dispersion. Linked survey data show the margins along which same-aged incumbents drifted apart: firms came to differ more in their innovation effort and in the number of products they sell, whereas the dispersion of sales per product did not change.

I embed permanent productivity heterogeneity into a quality-ladder model of firm dynamics with entry and exit and ask how much of the observed changes a single widening productivity gap can explain. Holding all other parameters at their calibrated baseline, a modest rise in the productivity gap reproduces the rise in within-age dispersion among older firms and most of the shift in the firm-age distribution. The mechanism is differential expansion: more productive firms expand faster, so size differences cumulate with tenure, raising dispersion among older firms while leaving young-firm dispersion flat. Fitted to the same targets, lower expansion costs or higher entry costs — natural alternatives from the literature on declining business dynamism — raise dispersion at all ages and steepen the mean size–age profile, counter to the data. A widening gap in R&D efficiency between the types reproduces the age signature, but not the decline in the young-firm share, and it raises rather than lowers growth. The productivity slowdown and the widening of productivity dispersion, which the Swedish data show directly and among young cohorts as much as overall, single out the productivity gap.

The widening gap reallocates market share toward more productive firms, raising the level of aggregate productivity, while slowing long-run growth mildly through a decline in entry. These forces largely offset along the transition between balanced growth paths: accounting for it, the welfare cost is only 0.042% in consumption-equivalent terms, far below the 1.92% implied by comparing balanced growth paths alone, which imposes the slowdown at once and misses the level gain from reallocation.

Several directions for future work are worth noting. First, the data span two decades. I interpret their start and end as balanced growth paths, although the transition between them takes longer than twenty years, so that the two balanced growth paths plausibly lie further apart in time than the sample. Documenting the trends over a longer period would allow the calibration to respect this distance. Second, the trends documented here closely align with those found for the US by Hopenhayn et al. (2022) and Karahan et al. (2024) on the margins they study, suggesting the dominance of within-age dispersion may be a broader feature of advanced economies; a natural next step is to apply the decomposition to other countries. Third, the same model-free decomposition applies to the *level* of firm-size dispersion observed in the data, not only to its change: in Kondziella (2026), I show that almost all firm-size dispersion in the cross-section is accounted for by dispersion conditional on age, with size differences across firm ages contributing negligibly — underscoring the importance of within-age heterogeneity for the level of dispersion as well.

References

Acemoglu, D., Akcigit, U., Alp, H., Bloom, N. and Kerr, W. (2018), 'Innovation, reallocation, and growth', *American Economic Review* **108**(11), 3450–91.

Aghion, P., Bergeaud, A., Boppart, T., Klenow, P. J. and Li, H. (2023), 'A theory of falling growth and rising rents', *Review of Economic Studies* **90**(6), 2675–2702.

Aghion, P., Bergeaud, A., Boppart, T., Klenow, P. J. and Li, H. (2025), Good rents versus bad rents: R&d misallocation and growth, Technical report, Centre for Economic Policy Research.

Akcigit, U. and Ates, S. T. (2021), 'Ten facts on declining business dynamism and lessons from endogenous growth theory', *American Economic Journal: Macroeconomics* **13**(1), 257–98.

Akcigit, U. and Ates, S. T. (2023), 'What happened to us business dynamism?', *Journal of Political Economy* **131**(8), 2059–2124.

Andrews, D., Criscuolo, C. and Gal, P. N. (2016), 'The best versus the rest: the global productivity slowdown, divergence across firms and the role of public policy'.

Autor, D., Dorn, D., Katz, L. F., Patterson, C. and Van Reenen, J. (2020), 'The fall of the labor share and the rise of superstar firms', *The Quarterly Journal of Economics* **135**(2), 645–709.

Berlingieri, G., Blanchenay, P. and Criscuolo, C. (2017), 'The great divergence (s)'.

Berlingieri, G., De Ridder, M., Lashkari, D. and Rigo, D. (2024), 'Growth through innovation bursts'.

Blundell, R., Griffith, R. and Windmeijer, F. (2002), 'Individual effects and dynamics in count data models', *Journal of econometrics* **108**(1), 113–131.

Cao, D., Hyatt, H. R., Mukoyama, T. and Sager, E. (2022), 'Firm growth through new establishments'.

Chen, Z. (2022), 'Economic growth and the rise of large firms'.

De Loecker, J., Eeckhout, J. and Unger, G. (2020), 'The rise of market power and the macroeconomic implications', *The Quarterly Journal of Economics* **135**(2), 561–644.

De Ridder, M. (2024), 'Market power and innovation in the intangible economy', *American Economic Review* **114**(1), 199–251.

De Ridder, M., Grassi, B. and Morzenti, G. (2026), 'The hitchhiker's guide to markup estimation: assessing estimates from financial data', *Econometrica* **94**(1), 137–168.

Decker, R. A., Haltiwanger, J., Jarmin, R. S. and Miranda, J. (2016*a*), 'Declining business dynamism: What we know and the way forward', *American Economic Review* **106**(5), 203–07.

Decker, R. A., Haltiwanger, J., Jarmin, R. S. and Miranda, J. (2016*b*), 'Where has all the skewness gone? the decline in high-growth (young) firms in the us', *European Economic Review* **86**, 4–23.

Decker, R. A., Haltiwanger, J., Jarmin, R. S. and Miranda, J. (2020), 'Changing business dynamism and productivity: Shocks versus responsiveness', *American Economic Review* **110**(12), 3952–90.

Engbom, N. (2023), 'Misallocative growth', *Working Paper* .

Gabaix, X. (2009), 'Power laws in economics and finance', *Annu. Rev. Econ.* **1**(1), 255–294.

Gourio, F., Messer, T. and Siemer, M. (2014), A missing generation of firms? aggregate effects of the decline in new business formation, *in* 'CES Ifo Conference'.

Grullon, G., Larkin, Y. and Michaely, R. (2019), 'Are us industries becoming more concentrated?', *Review of Finance* **23**(4), 697–743.

Hall, B. H. and Ziedonis, R. H. (2001), 'The patent paradox revisited: an empirical study of patenting in the us semiconductor industry, 1979-1995', *rand Journal of Economics* pp. 101–128.

Hopenhayn, H., Neira, J. and Singhania, R. (2022), 'From population growth to firm demographics: Implications for concentration, entrepreneurship and the labor share', *Econometrica* **90**(4), 1879–1914.

Hottman, C. J., Redding, S. J. and Weinstein, D. E. (2016), 'Quantifying the sources of firm heterogeneity', *The Quarterly Journal of Economics* **131**(3), 1291–1364.

Karahan, F., Pugsley, B. and Şahin, A. (2024), 'Demographic origins of the start-up deficit', *American Economic Review* **114**(7), 1986–2023.

Klette, T. J. and Kortum, S. (2004), 'Innovating firms and aggregate innovation', *Journal of political economy* **112**(5), 986–1018.

Kondziella, M. (2026), 'Firm-size dispersion and the limits of age'.

Kwon, S. Y., Ma, Y. and Zimmermann, K. (2024), '100 years of rising corporate concentration', *American Economic Review* **114**(7), 2111–2140.

Lentz, R. and Mortensen, D. T. (2008), 'An empirical model of growth through product innovation', *Econometrica* **76**(6), 1317–1373.

Liu, E., Mian, A. and Sufi, A. (2022), 'Low interest rates, market power, and productivity growth', *Econometrica* **90**(1), 193–221.

Luttmer, E. G. (2007), 'Selection, growth, and the size distribution of firms', *The Quarterly Journal of Economics* **122**(3), 1103–1144.

Luttmer, E. G. (2011), 'On the mechanics of firm growth', *The Review of Economic Studies* **78**(3), 1042–1068.

Paron, J. D. (2026), 'The wealth of stagnation: Falling growth, rising valuations', *Rising Valuations (January 19, 2026)* .

Peters, M. and Walsh, C. (2021), Population growth and firm dynamics, Technical report, National Bureau of Economic Research.

Supplemental Appendix for

"Recent Changes in Firm Dynamics and the Nature of Rising Firm-Size Dispersion"

[FOR ONLINE PUBLICATION]

A Data

The main dataset, *Företagens Ekonomi* (FEK), contains information from balance sheets and profit and loss statements for the universe of Swedish firms. From this dataset, I obtain the firm-size measures, including sales, intermediate inputs, the capital stock, employment, and the wage bill. In the FEK code-book by Statistics Sweden (SCB), these variables are defined as follows.[27] SCB defines sales as income from the companies' main business from goods sold and services provided. As the measure of the firm's capital stock I use fixed assets minus depreciation. Employment refers to the average number of employees in full-time units in accordance with the company's annual report. The measure of the firm's wage bill includes salaries and other remuneration, including severance pay. As described in the main text, I focus on firms in the private sector. These firms have a legal type (variable name: *JurForm*) less than 50 or equal to 96.

Starting in 2004, the data include unincorporated self-employed with negative profits. These firms were previously not recorded, leading to an inflow of new firms in that year. Since I cannot differentiate between firms established in 2004 and firms entering the data in 2004 with an earlier birth year, I drop firms that enter the data in 2004 with negative profits when showing the firm-age distribution. When showing the size-by-age patterns, I leave the 2004 cohort out.[28] For the main exercise — which decomposes the change in size dispersion between 2002 and 2017 — this is inconsequential: firm ages are grouped, so firms born in 2004 fall into the highest age group in 2017 regardless.

[27] `https://www.scb.se/contentassets/9dd20ce462644cc19f6f04eb2edbbe28/nv0109_kd_slut2022_v1_20240510.pdf`, accessed 26.03.2025.

[28] Leaving the 2004 cohort out when computing the age distribution would bias it in all years following 2004.

To measure size dispersion within sectors, I use the industry classification (SNI codes) provided by SCB, which changed in 2002 and in 2007. The changes mainly affect the more detailed codes at the five-digit level, whereas I use the one-digit sections for the sector breakdowns of Section 4 and two-digit industries in Section 7 and Table A-1. To obtain a consistent classification over time, I extend the new classification back in time. In the year of a change, both classifications are observed, so the codes of almost all firms can be updated directly. For a firm absent from the data in that year but present one year after the change, I use the classification observed after the change; for firms absent for several years around the change, I use the industry mappings provided by SCB and, where a mapping is many-to-many, the most common transition.

One concern is that changes in the firm structure, e.g., when firms merge with other firms, change the firm ID. To address this concern, I impute changes in firm IDs using worker flows between firms. The auxiliary data set *Registerbaserad Arbetsmarknadsstatistik* (RAMS) contains the universe of employer-employee matches. I impute changes in the firm ID of firms with at least five employees as follows: if more than 50% of the workforce of firm A in year t makes up more than 50% of the workforce of firm B in year $t+1$, I replace firm B's firm ID with firm A's firm ID from $t+1$ onward. The empirical results remain unchanged when excluding firms for which the imputed firm ID differs from the observed firm ID.

Survey of industrial production (IVP)

The survey of industrial production (*Industrins varuproduktion*) records, for each manufacturing firm and year, the value and quantity of production delivered by product code. Product codes follow the eight-digit Combined Nomenclature, refined by Statistics Sweden with additional subdivisions; separate codes cover industrial services, and a flag distinguishes own production from contract work. The survey is a census with a cut-off: firms with at least 20 employees are surveyed (10 in a few industries, and from 2006 also firms with turnover above 75 million SEK); production of smaller firms and of non-respondents is imputed by an allocation model that distributes sales over codes, and the delivered data flag imputed values. I use the years 1997–2017 and keep surveyed responders only, dropping firm-years in which all values are imputed, since their product structure is an artifact of the imputation; I further drop contract work and the industrial-service codes, so that a firm's products

are the goods it produces itself, and require at least 20 employees in FEK in every year, because the survey threshold was 10 employees until 2002 and 20 thereafter. A firm's number of products is the number of distinct six-digit codes with positive production value; counting at six rather than eight digits mutes the annual revisions of the eight-digit nomenclature. Between 2003 and 2006 part of the surveyed firms were rotated biennially, so the set of responders is smaller in those years. The resulting sample contains 2,100–3,100 firm-years per year, 97–99% of which are linked to FEK on the firm identifier. Writing a firm's production value as the number of products times sales per product, $\ln S = \ln n + \ln(S/n)$, the variance of log production value is the sum of the variances of the two terms and twice their covariance; Figure 5 reports the standard deviations of the two terms.

Business R&D survey (FoU)

The business R&D survey (*Forskning och utveckling i företagssektorn*) is conducted every second year; I use the ten waves 1999–2017. Its frame consists of firms with at least ten employees from 2005 onward and at least 50 before; firms with 200 or more employees, firms in the R&D-services industry, and firms that reported more than five million SEK of R&D in the previous wave are surveyed in full, smaller firms are sampled. Amounts are reported in thousands of nominal SEK for the calendar year. I use a firm's total expenditure on R&D performed in-house: labor cost of R&D staff, fees of consultants working under the firm's direction, other operating costs, and investment in buildings, machinery, and software for R&D. Firms without R&D return the form and are recorded with zero or blank expenditure. Respondents are linked to FEK on the firm identifier (82–95% per wave). I keep respondents with at least 50 employees in FEK, the size class inside the frame in every wave, and, for the dispersion measures, those with positive own R&D; this gives 540–700 R&D-performing firms per wave, 400–570 of them aged 11 or older. One check supports the dispersion series in Figure 5: smaller performers are sampled rather than surveyed in full, so each responding firm stands for several in the population; recomputing the standard deviation with the survey's design weights, available through 2013, gives the same or a larger rise (1.86 in 1999 to 2.33 in 2013, against 1.85 to 2.13 unweighted). R&D intensity divides R&D by net sales of the same year, both in nominal terms; within-year dispersion in logs is in any case invariant to deflation.

Innovation activity, size, and growth in the linked data

The linked data let me verify, in the samples described above, the basic correlations on which a model of firm growth through innovation rests. Among R&D performers, R&D expenditure scales roughly proportionally with sales (an elasticity of 0.82; 0.92 in the fully surveyed stratum of firms with 200 or more employees), and R&D intensity is highly persistent within firms, with a coefficient of 0.80 from one survey wave to the next. Firms that perform R&D grow faster and exit less over the following two years than non-performers of the same size, industry, and age. Among performers, higher R&D intensity predicts faster sales growth and more product additions over the following two to four years, and adding a product coincides with hiring in the same year. The first three are among the stylized facts that motivated the model class (Klette and Kortum, 2004); the last two — that a firm's R&D predicts its own subsequent product additions, and that adding a product goes together with hiring — require observing R&D and product scope for the same firms, which the linked surveys make possible. Table A-1 reports the estimates. Each row is a separate regression of the outcome on the regressor within cells defined by year, two-digit industry, and firm-age bin (0–5, 6–10, 11 and older), with standard errors clustered by firm; cells with a single observation are dropped, so the reported number of observations counts those that identify the coefficient. Growth outcomes are log differences for firms with positive values in both years; the Davis–Haltiwanger–Schuh (DHS) growth rate, $2(x_{t+h} - x_t)/(x_{t+h} + x_t)$, is set to -2 for firms without a record h years ahead and so combines growth and exit; exit is the absence of a record h years ahead. Sales growth is measured in 2017 SEK, R&D intensity in nominal terms of the same year.

Other data sources

Aggregate TFP growth comes from Federal Reserve Economic Data (FRED): Total Factor Productivity at Constant National Prices for Sweden (series RTFPNASEA632NRUG), accessed 30.01.2024.

B Size dispersion and firm dynamics in the data

Table A-1: Innovation activity, size, and growth in the linked data

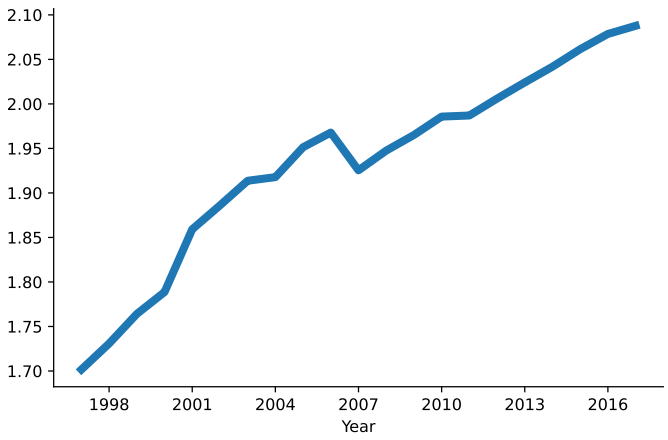
Outcome	Regressor	Sample	N	Coef. (SE)
Levels				
ln R&D	ln sales	R&D performers, $\geq$ 200 empl.	3,011	0.921 (0.059)
ln R&D	ln sales	R&D performers, $\geq$ 50 empl.	5,552	0.823 (0.037)
ln number of products	ln sales	IVP	54,177	0.191 (0.009)
Persistence				
ln R&D intensity, $t+2$	ln R&D intensity, t	R&D performers, $\geq$ 50 empl.	3,397	0.798 (0.020)
Growth and survival, t to $t+2$				
DHS employment growth	R&D performer (0/1)	FoU respondents, $\geq$ 50 empl.	11,871	0.034 (0.013)
Exit	R&D performer (0/1)	FoU respondents, $\geq$ 50 empl.	11,871	$-$0.013 (0.004)
Sales growth	ln R&D intensity	R&D performers, $\geq$ 50 empl.	4,847	0.020 (0.009)
Employment growth	ln R&D intensity	R&D performers, $\geq$ 50 empl.	4,823	$-$0.000 (0.006)
Product-count growth	ln R&D intensity	R&D performers in IVP	2,959	0.013 (0.006)
Products and hiring, t to $t+1$				
Employment growth	Any product added (0/1)	IVP	43,441	0.027 (0.004)

Notes: Each row is a separate regression within year $\times$ two-digit industry $\times$ age-bin cells, standard errors clustered by firm. The two rows on R&D performers versus non-performers and the sales- and employment-growth rows control for ln sales at t; the product-addition row controls for the ln number of products at t; the remaining rows have no controls. Firms with 200 or more employees are surveyed in full, so the first row is free of the survey's sampling design; the second row is unweighted. R&D intensity is own R&D expenditure over net sales. DHS employment growth sets exits to -2; exit is the absence of a firm record two years ahead. At a four-year horizon the coefficients are 0.034 (0.020) for DHS employment growth, $-$0.016 (0.007) for exit, 0.045 (0.014) for sales growth, 0.004 (0.010) for employment growth, and 0.024 (0.010) for product-count growth. The product-addition row conditions on the firm being surveyed in $t+1$; its coefficient is unchanged among firms with at least 50 employees, comfortably above the survey threshold. One-year changes in the number of products and in employment are correlated within the year (0.09). Samples as described above; 2-year horizons correspond to consecutive R&D survey waves.

B.1 Firm-size dispersion

Figure A-1 restricts to firms with positive sales and plots the standard deviation of log sales. The standard deviation increases by about 0.4 log points.

Figure A-1: Firm-size dispersion (standard deviation)

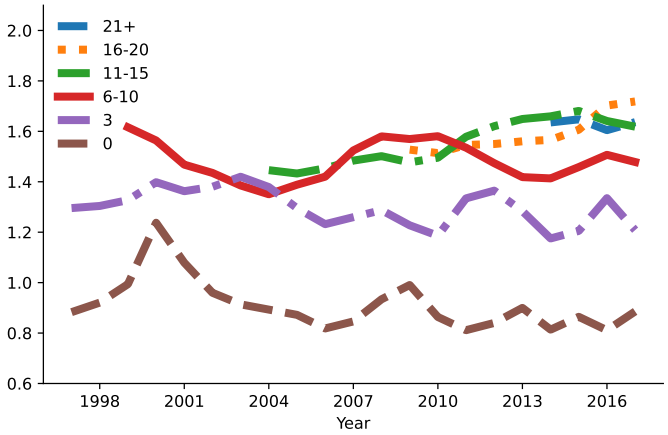


Notes: The figure shows the standard deviation of log sales, restricting to firm-year observations with positive sales.

B.2 Firm size conditional on age

Figure A-2 shows the log of the average firm size by age, where size is measured using employment. There are no systematic trends, including among entrants.

Figure A-2: Firm size conditional on age (employment)



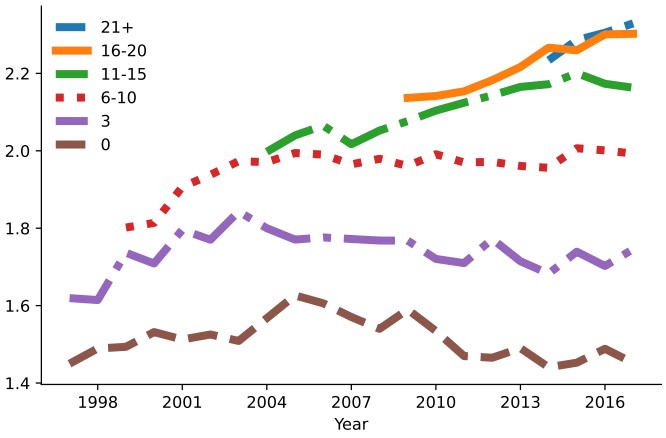
Notes: The figure shows the log of the average firm size (employment) by firm age indicated in the legend.

B.3 Firm-size dispersion conditional on age

As a first robustness check, I use the standard deviation instead of the IQR. This is unproblematic for sales since few firms report zero sales, so restricting to positive observations is not consequential. For other size measures such as employment or the wage bill, restricting to positive values is more consequential and introduces stronger selection — self-employed workers, for instance, commonly report a zero wage bill.

As shown in Figure A-3, restricting to firms with positive sales and using the standard deviation of log sales as the measure of dispersion leaves the trends unchanged. Size dispersion conditional on age remains relatively stable for young firms and increases by about 0.2 log points for the age groups 6–10, 11–15 and 16–20.

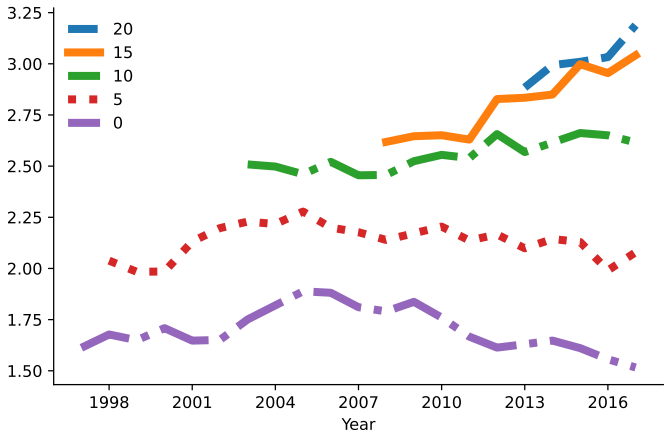
Figure A-3: Size dispersion conditional on age (standard deviation)



Notes: The figure shows the standard deviation of log sales by firm age as indicated in the legend. The figure restricts to firm-year observations with positive sales.

I next condition on single ages instead of age groups. Figure A-4 shows size dispersion for firms aged zero, five, ten, fifteen and twenty. The patterns remain the same: size dispersion shows no trend among young firms, but increased systematically among old firms.

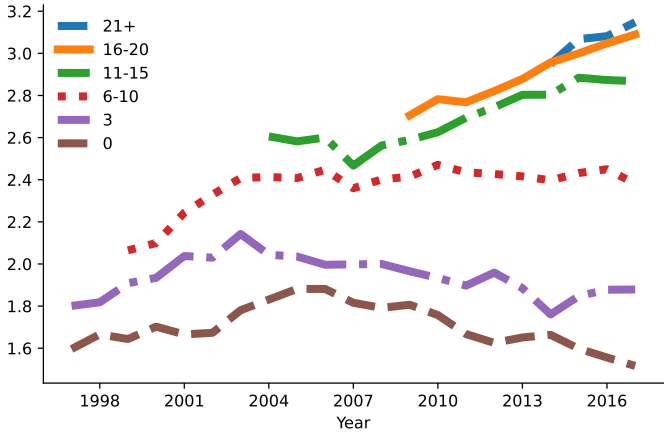
Figure A-4: Size dispersion conditional on age (alternative ages)



Notes: The figure shows the interquartile range (IQR) defined as the difference between the 75th and 25th percentile of the log sales distribution by firm age as indicated in the legend.

The picture also remains unaffected when firm sales are deflated to 2017 SEK with sector-level GDP deflators instead of the aggregate GDP deflator (Figure A-5). Finer deflators leave size dispersion within sectors unchanged and could affect it only at the aggregate level; the figure shows that they do not: differential inflation across sectors has no effect on the change in log sales dispersion.

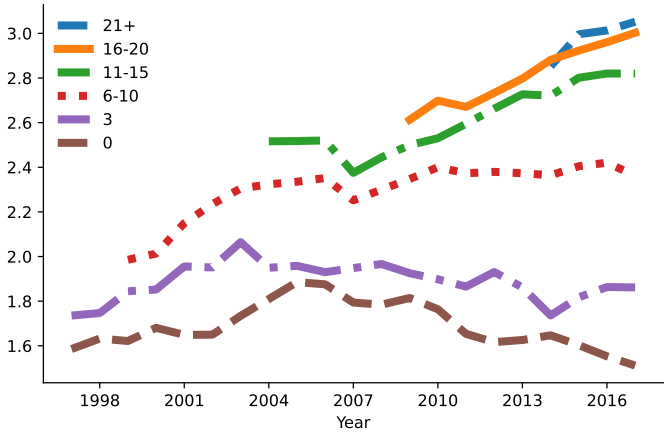
Figure A-5: Size dispersion conditional on age (sector deflators)



Notes: The figure shows the interquartile range (IQR) defined as the difference between the 75th and 25th percentile of the log sales distribution by firm age as indicated in the legend. Sales are deflated using sector-specific GDP deflators.

Mergers and acquisitions have not contributed to the rise either. In the data, any change in firm structure results in a new firm ID; Supplemental Appendix A describes how I identify such changes from worker flows and link the firm IDs. Figure A-6 excludes the firms for which the observed firm ID differs from the imputed one; the picture is virtually unchanged.

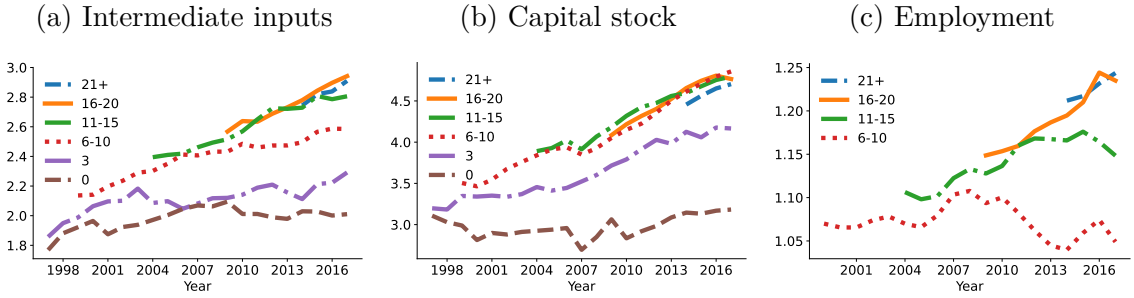
Figure A-6: Size dispersion conditional on age (excluding mergers)



Notes: The figure shows the interquartile range (IQR) defined as the difference between the 75th and 25th percentile of the log sales distribution by firm age as indicated in the legend. The figure excludes firms with an imputed firm ID change.

Next, I assess the robustness of the increase in firm-size dispersion conditional on age for three alternative size measures: intermediate inputs, the capital stock, and employment. Few firms report zero intermediate inputs, so the IQR applies as for sales. The capital stock and employment require a different measure. The 25th percentile of the capital stock is commonly zero (Table 1), which leaves the IQR of log capital undefined; I therefore show the difference between the 95th and the 50th percentile of the log capital stock, a measure of dispersion in the right tail. The 25th percentile of employment is zero as well, since the self-employed hire no workers, and employment is a very discrete measure of size for most firms (the median firm employs one worker), so the IQR does not accurately reflect changes in its dispersion; for employment, I therefore exclude firms with zero employment and use the standard deviation.

Figure A-7: Size dispersion conditional on age (alternative size measures)

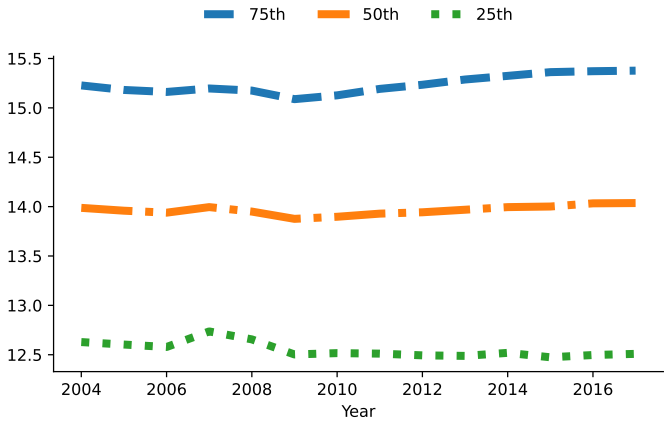


Notes: The figure shows the dispersion of log firm size by firm age as indicated in the legend for alternative size measures. The left panel uses intermediate inputs and applies the interquartile range as the dispersion measure. The middle panel uses the capital stock and shows the difference between the 95th and 50th percentile. The right panel uses employment and applies the standard deviation as the dispersion measure (restricting to positive values).

Figure A-7 shows firm-size dispersion conditional on age for the three alternative size measures. The increase in size dispersion conditional on age is even more visible when using intermediate inputs, and is again particularly pronounced for older firms: within-group size dispersion for the age groups 6–10, 11–15, and 16–20 increases by about 0.4 log points. The increase is even stronger when using the capital stock as the size measure: for the same age groups the increase exceeds 0.5 log points. This should, however, be interpreted with caution, as the measure focuses on dispersion in the right tail of the size distribution. Moreover, since the capital stock comprises only tangible assets, the divergence extends to physical capital and is not confined to intangibles. The last panel shows dispersion in log employment conditional on age over time for firms with at least one worker. Since this restriction heavily selects the sample, particularly among young firms, I only show the trends for older firm ages. For the age groups 11–15 and 16–20, a clear upward trend is noticeable. This increase is slightly smaller than the increase in sales dispersion shown for the standard deviation in Figure A-3, though sample selection again calls for caution.

Next, I show the size distribution conditional on age to examine how individual percentiles have shifted. Figure A-8 shows the 25th, 50th, and 75th percentile of the log sales distribution for 11–15 year old firms. The IQR widens at both ends: between 2004 and 2017, the 25th percentile fell by 0.12 log points and the 75th percentile rose by 0.15.

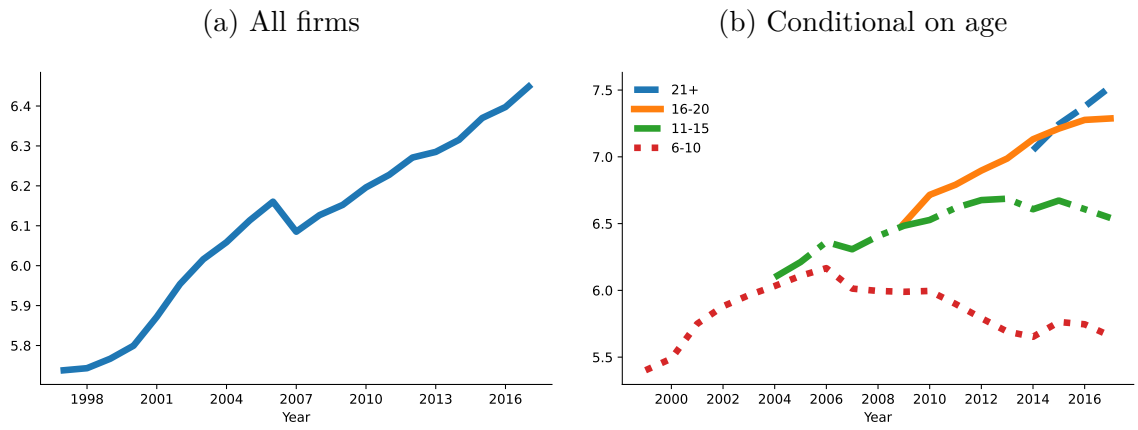
Figure A-8: Size distribution (ages 11–15)



Notes: The figure shows percentiles (25th, 50th, 75th) of the log sales distribution for 11–15 year old firms.

Lastly, I document the extreme right tail of the size distribution directly. The dispersion measures used so far describe the bulk of the distribution, and sales are winsorized at the 0.1 and 99.9 percent levels. Figure A-9 replaces the 75th percentile in the interquartile range by the 99th percentile, i.e., it shows the difference between the 99th and the 25th percentile of log sales, and it uses unwinsorized sales. Keeping the 25th percentile as the lower bound makes the figure directly comparable to Figures 1 and 4 in the main text: any difference between them comes from the 99th percentile in place of the 75th. Panel (a) shows the measure for all firms; panel (b) shows it conditional on age for firms aged six and older. The youngest age groups of Figure 4, ages zero and three, are omitted because a misclassification of firm age would show up in the right tail of the young-firm size distribution. Excluding firms with an imputed change in firm structure leaves the interquartile range unaffected (Figure A-6), but the imputation from worker flows is an imperfect filter, and the 99th percentile of young firms in unwinsorized data would be more sensitive to the cases it misses.

Figure A-9: Firm-size dispersion in the extreme right tail



Notes: The figure shows the difference between the 99th and the 25th percentile of the log sales distribution, computed on unwinsorized sales. Panel (a) shows the measure for all firms; panel (b) shows it by firm age as indicated in the legend, for firms aged six and older.

The trends documented in the main text are also present in the extreme right tail, and they are more pronounced there. For all firms, the gap between the 99th and the 25th percentile widens by 0.7 log points between 1997 and 2017, compared with 0.5 log points for the interquartile range. Conditional on age, the rise is again concentrated among older firms. For the age groups 11–15, 16–20, and 21+, the right-tail gap widens by 0.4 to 0.8 log points over the years in which each group is observed, compared with 0.2 to 0.4 log points for the interquartile range over the same years. For firms aged 6–10, the gap widens until 2006 and narrows again thereafter; its net increase over 1999–2017 is about 0.3 log points, the same as for the interquartile range.

B.4 Alternative mechanisms

This subsection takes up three prominent alternative explanations of the rise in size dispersion conditional on age. Each fails a qualitative test: it implies a footprint across firm ages, sectors, or the two tails of the size distribution that the data of Section 4 do not show. The key discriminating fact is the age profile of the rise, which points to forces that accumulate with firm tenure.

First, weakening selection among unproductive firms — for instance, due to falling in-

terest rates keeping otherwise non-viable firms alive — predicts that the effect should be strongest in capital-intensive sectors reliant on debt financing, and that it should primarily inflate the left tail of the size distribution by keeping small firms alive. Both predictions fail. The rise in size dispersion conditional on age is weakest precisely in Manufacturing (Figure 4), the most capital-intensive sector in the data. Moreover, as shown separately for each tail in Figure A-8, the 75th percentile of log sales rose while the 25th percentile fell over the observed time period. The simultaneous expansion of both tails is more consistent with genuine productivity divergence than with zombie-firm survival.

Second, rising size dispersion conditional on age could reflect changes in the composition of entrants over time rather than post-entry divergence among incumbents. Figure 4 rules this out: size dispersion among young firms shows no trend over the sample period, unlike dispersion among older firms, implying that divergence accumulates after entry rather than being inherited from entry conditions.[29] Young cohorts did become more dispersed in productivity (Section 7), but not in size: the size differences appear only as the productivity differences cumulate with tenure, as the model of Section 5 implies.

Third, the same age profile rules out explanations based on noise common to firms of all ages, such as a rise in measurement error. Because such noise is realized independently of firm tenure, it would inflate within-age dispersion uniformly across ages — raising it among entrants and young firms as much as among older firms. The data show the opposite: the rise is concentrated among older firms and absent among the young, the signature of persistent differences that accumulate over a firm's life rather than of age-invariant noise.[30]

C The mechanism in the data

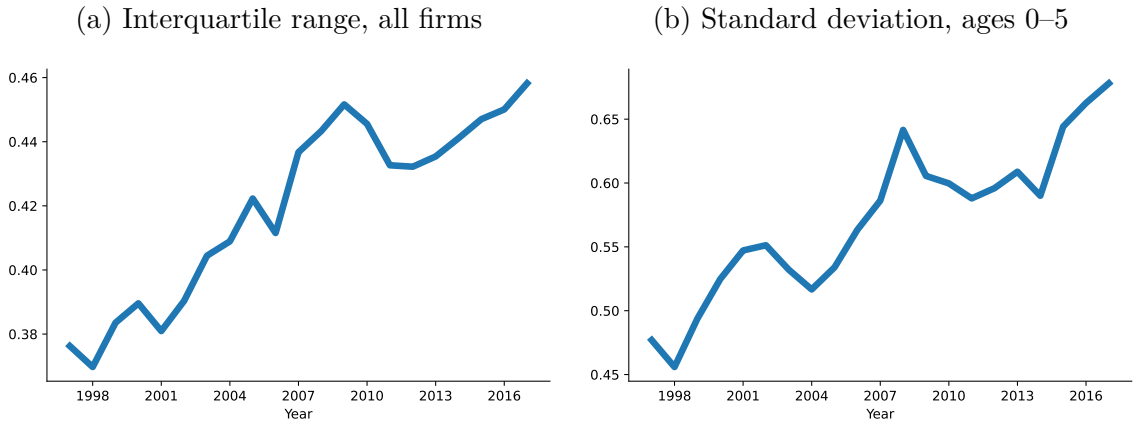
Figure A-10 complements Figure 10 in the main text. Panel (a) replaces the standard deviation by the interquartile range of log revenue productivity within two-digit in-

[29] One might worry that mean entrant quality has declined over time, even if dispersion among entrants is stable. Average sales among entrants did decline slightly (Figure 2), but average employment among entrants shows no systematic trend (Supplemental Appendix B.2), suggesting that the real size of entrants did not meaningfully change.

[30] The pervasiveness of the rise in size dispersion conditional on age across service sectors (Figure 4) — whose output is largely non-tradable — further rules out foreign demand as a driver.

dustries, computed and averaged across industries in the same way. Panel (b) shows the standard deviation for firms aged zero to five.

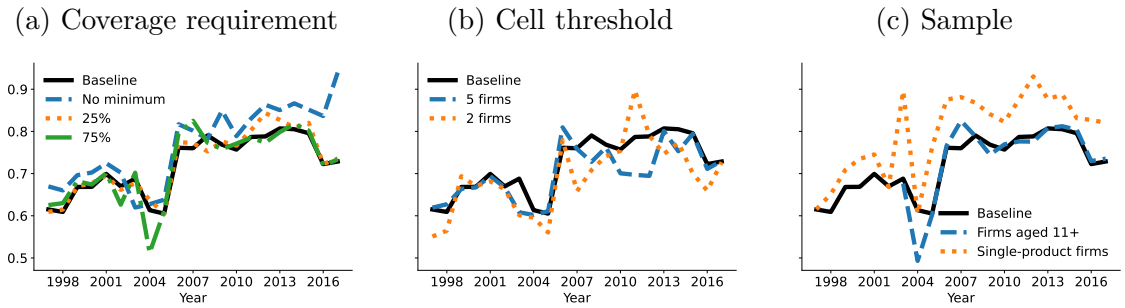
Figure A-10: Dispersion of revenue productivity within industries (robustness)



Notes: Log revenue productivity is defined in eq. (16). Both statistics are computed for each two-digit industry and year (cells with at least ten firms) and averaged across industries with the industries' total costs as weights. Panel (a) shows the interquartile range for all firms. Panel (b) shows the standard deviation for firms aged zero to five.

Figure A-11 complements Figure 11 in the main text. Each panel repeats the baseline series of physical productivity dispersion among surveyed manufacturers and changes one element of the construction that the main text holds fixed: the share of a firm's production value that the products with observed quantities must cover (panel a), the minimum number of firms per industry-year cell (panel b), and the sample, with firms aged eleven and older, shown from 2003 when this age group can be identified, and single-product firms, whose price is a single relative price (panel c). As in the main text, the level of dispersion and the size of its increase vary with the construction; its direction does not.

Figure A-11: Dispersion of physical productivity among surveyed manufacturers (robustness)



Notes: The baseline series is that of Figure 11. Panel (a) varies the minimum share of a firm's production value that its products with observed quantities must cover (none, 25, or 75 percent instead of 50). Panel (b) lowers the minimum number of firms per two-digit industry and year cell from ten to five or two. Panel (c) restricts the sample to firms aged eleven and older or to firms with at least 90 percent of their production value in one product.

D Model

D.1 Household and static allocation

This section states the household problem and derives the static allocation summarized in Section 5, taking the joint distribution of product qualities and firm productivity as given.

Household The representative household chooses the path of consumption C_t and wealth A_t to maximize lifetime utility

$$U = \int_0^\infty \exp(-\rho t) \ln C_t dt,$$

subject to the budget constraint $\dot{A}_t = r_t A_t + w_t L_t - C_t$. ρ denotes the discount rate, r_t the interest rate and w_t the real wage. The household supplies one unit of labor inelastically, i.e., $L_t = 1$. The optimality condition (Euler equation) for the household problem reads

$$\frac{\dot{C}_t}{C_t} = r_t - \rho.$$

Product level Expenditure minimization in the production of the final good leads to the demand function

$$y_i = \beta_i \frac{YP}{p_i}, \tag{A-1}$$

where P is defined as the aggregate price index, which I normalize to 1. Firms in product market i compete in prices (Bertrand competition). In equilibrium, only the firm with the highest quality-adjusted productivity $q_{if}\varphi_f$ produces product i (henceforth, incumbent). Under Bertrand competition, the incumbent firm engages in limit pricing and sets its price equal to the quality-adjusted marginal costs of the follower (the firm with the second highest quality-adjusted productivity)

$$p_{if} = \frac{q_{if}}{q_{if'}} \frac{w}{\varphi_{f'}}, \tag{A-2}$$

where f' indexes the follower in product market i. The price that the incumbent sets is increasing in the quality gap between the incumbent and the follower, as eq. (A-2) shows. Dividing the price by the marginal costs of the incumbent, w/φ_f, yields the product markup in eq. (2), which is increasing in the quality and productivity gap of the incumbent. The price setting of the incumbent gives rise to the following profits for product i

$$\pi_{if} = p_{if} y_{if} - w l_{if} = \beta_i Y \left(1 - \frac{1}{\mu_{if}}\right),$$

with labor demand for product i

$$l_{if} = \beta_i \frac{Y}{w} \mu_{if}^{-1}. \tag{A-3}$$

Sales of product i equal $\beta_i Y$. Profits and employment in product line i scale with its weight, and employment is decreasing in the markup.

Firm level Firm employment is the sum of employment across the firm's product lines

$$l_f = \sum_{i \in N_f} l_{if} = \frac{Y}{w} \left(\sum_{i \in N_f} \beta_i \mu_{if}^{-1} \right),$$

where N_f denotes the set of product lines in which firm f is the incumbent producer. Firm employment decreases in the markups of the firm's product lines and increases in their number. Firm sales are given by eq. (3).

Aggregate level Integrating employment across firms or products yields the total workforce in production:

$$L_P = \int_f l_f df = \frac{Y}{w} \int_0^1 \beta_i \mu_{if}^{-1} di = \frac{Y}{w} \int_0^1 \mu_{if}^{-1} di. \tag{A-4}$$

The last equality holds because the weights have mean one and are independent of markups across products. The same argument applies to every aggregate below: an expenditure-weighted average across products, $\int_0^1 \beta_i h_i di$, of any product characteristic h_i that is independent of the weights equals the unweighted average $\int_0^1 h_i di$. The weights thus drop out of all aggregates, and I omit them in the expressions that follow. Taking logs and integrating eq. (2), one obtains an expression for the wage

$$w = \exp\left(\int_0^1 \ln q_{if} di\right) \times \exp\left(\int_0^1 \ln \varphi_{f(i)} di\right) \times \exp\left(\int_0^1 \ln \mu_{if}^{-1} di\right). \tag{A-5}$$

Inserting eq. (A-5) into eq. (A-4) yields the expression for aggregate output in eq. (4), $Y = Q\Phi\mathcal{M}L_P$.

D.2 Solving the dynamic firm problem

Denoting by $[\mu_i, \beta_i]$ a firm's set of markups and weights across its product lines, firm profits follow

$$\pi_f(n, [\mu_i, \beta_i]) \equiv \sum_{k=1}^{n} \pi_i(\mu_k, \beta_k) = \sum_{k=1}^{n} \beta_k Y \left(1 - \frac{1}{\mu_k}\right) = \sum_{k=1}^{n} \beta_k Y \left(1 - \frac{1}{\lambda} \frac{1}{\frac{\varphi_{f(k)}}{\varphi_{f'(k)}}}\right).$$

For their R&D activities, firms pay a cost of

$$\Gamma\left([x_{it}]; n\right) = \sum_{k=1}^{n} c\left(x_{kt}\right) = \sum_{k=1}^{n} \frac{1}{\psi_x} (x_{kt})^{\zeta}$$

in terms of labor.

Incumbent firms choose the rate of expansion R&D, x_{it}, for each of their product

lines, i. When choosing x_{it}, firms take aggregate output Y_t, the real wage w_t, the share of lines operated by high-productivity firms S_t, the interest rate r_t and the rate of creative destruction τ_t as given. Note that S_t is predetermined in t; its law of motion is eq. (7). Denoting the time derivative by $\dot{V}_t^h()$, the value function of a high-productivity type firm (indexed by h) satisfies the following HJB equation:

$$
\begin{aligned}
&r_t V_t^h(n,[\mu_i,\beta_i],S_t) - \dot{V}_t^h(n,[\mu_i,\beta_i],S_t) = \\
&\quad \sum_{k=1}^{n} \underbrace{\pi_i(\mu_k,\beta_k)}_{\text{Flow profits}} + \sum_{k=1}^{n} \underbrace{\tau_t \Big[V_t^h\left(n-1,[\mu_i,\beta_i]_{i\neq k},S_t\right) - V_t^h(n,[\mu_i,\beta_i],S_t) \Big]}_{\text{Creative destruction}} \\
&\quad + \max_{[x_{it}]} \Bigg\{ \underbrace{\sum_{k=1}^{n} x_{kt} \Big[S_t E_{\beta'} V_t^h\left(n+1,[[\mu_i,\beta_i],(\lambda,\beta')],S_t\right) + (1-S_t) E_{\beta'} V_t^h\left(n+1,\left[[\mu_i,\beta_i],\left(\lambda\cdot\frac{\varphi^h}{\varphi^\ell},\beta'\right)\right],S_t\right) - V_t^h(n,[\mu_i,\beta_i],S_t) \Big]}_{\text{Expansion R\&D}} \\
&\quad - \underbrace{w_t \Gamma\left([x_{it}];n\right)}_{\text{R\&D costs}} \Bigg\}
\end{aligned}
$$

The value of a firm consists of flow profits, research costs, and two terms related to creative destruction and expansion R&D. At the rate of creative destruction τ_t (determined in equilibrium), the firm loses one of its n products, in which case it remains with $n-1$ products. At the optimally chosen rate x_{it}, expansion R&D is successful (third row), and the firm acquires a new product (n increases by one). Since innovation is undirected, the weight β' of the new product is a draw from G, over which the firm takes expectations, $E_{\beta'}$.

Productivity types enter the value function through the markup expectations. The share of product lines operated by high-productivity firms, S_t, is a state variable that firms keep track of to form them. When taking over a new product line through expansion R&D (third row), the probability of replacing a high-type incumbent is S_t, in which case the high-type firm charges a markup of λ. With probability $1-S_t$, the replaced incumbent is of the low type, and the high-type firm charges a markup of $\lambda \cdot \varphi^h/\varphi^\ell$.

The HJB equation for a low-productivity type firm (indexed by ℓ) follows the same structure. The term related to expansion R&D (third row) is distinct since markup expectations differ for low-productivity firms:

$$
\begin{aligned}
&r_t V_t^\ell(n,[\mu_i,\beta_i],S_t) - \dot{V}_t^\ell(n,[\mu_i,\beta_i],S_t) = \\
&\sum_{k=1}^{n}\pi_i(\mu_k,\beta_k) + \sum_{k=1}^{n}\tau_t\Big[V_t^\ell\left(n-1,[\mu_i,\beta_i]_{i\neq k},S_t\right) - V_t^\ell(n,[\mu_i,\beta_i],S_t)\Big] \\
&+ \max_{[x_{it}]}\Bigg\{\sum_{k=1}^{n}x_{kt}\Big[S_t E_{\beta'}V_t^\ell\left(n+1,\left[[\mu_i,\beta_i],\left(\lambda\cdot\frac{\varphi^\ell}{\varphi^h},\beta'\right)\right],S_t\right) + (1-S_t)E_{\beta'}V_t^\ell\left(n+1,[[\mu_i,\beta_i],(\lambda,\beta')],S_t\right) - V_t^\ell(n,[\mu_i,\beta_i],S_t)\Big] \\
&- w_t\Gamma\left([x_{it}];n\right)\Bigg\}
\end{aligned}
$$

The expected values of entering as a high- or low-type firm, which enter the free-entry condition (8), are

$$
\begin{aligned}
E[V_t^h(1,\mu_i,\beta_i)] =& S_t E_{\beta'}V_t^h(1,\lambda,\beta') + (1-S_t)E_{\beta'}V_t^h(1,\lambda\times\varphi^h/\varphi^\ell,\beta') \\
E[V_t^\ell(1,\mu_i,\beta_i)] =& S_t E_{\beta'}V_t^\ell(1,\lambda\times\varphi^\ell/\varphi^h,\beta') + (1-S_t)E_{\beta'}V_t^\ell(1,\lambda,\beta').
\end{aligned}
$$

Definition 1. *A balanced growth path (BGP) is a set of allocations* $[y_{it}, l_{it}, x_{it}, z_t, S_t, C_t]_t$ *and prices* $[r_t, w_t, p_{it}]_t$ *such that firms choose* $[x_{it}, p_{it}]_t$ *optimally, the representative household maximizes utility choosing* $[y_{it}, C_t]_t$*, the growth rate of aggregate variables is constant, the free-entry condition holds, all markets clear and the distribution of quality and productivity gaps* ν_t *is stationary.*

For clarity, I suppress the dependence of the value function on S_t in the following. Following a guess-and-verify approach, I obtain the value function of a firm of productivity type d with n products

$$
V_t^d(n,[\mu_i,\beta_i]) = n\frac{1}{(\rho+\tau)}\frac{\zeta-1}{\psi_x}(x^d)^\zeta w_t + \sum_{k=1}^{n}\frac{\pi_i(\mu_k,\beta_k)}{\rho+\tau}, \tag{A-6}
$$

where $\pi_i(\mu_k,\beta_k) = \beta_k Y_t(1-1/\mu_k)$. The value function is linear in the weight of each product. The expected value of a new product over its weight β', which enters the expansion term of the HJB equation, therefore equals the value at the mean weight, $E_{\beta'}V_t^d\left(n+1,[[\mu_i,\beta_i],(\mu',\beta')]\right) = V_t^d\left(n+1,[[\mu_i,\beta_i],(\mu',1)]\right)$, and the weights drop out of the optimality conditions. The optimality condition for expansion R&D for high-productivity firms is given by

$$
\frac{\zeta-1}{\psi_x}(x^h)^\zeta + S_t\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\right) + (1-S_t)\frac{Y_t}{w_t}\left(1-\frac{\varphi^\ell}{\varphi^h}\frac{1}{\lambda}\right) = (\rho+\tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta-1}
$$

and for low-productivity firms by

$$\frac{\zeta-1}{\psi_x}(x^\ell)^\zeta + S_t\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\frac{\varphi^h}{\varphi^\ell}\right) + (1-S_t)\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\right) = (\rho+\tau)\frac{\zeta}{\psi_x}(x^\ell)^{\zeta-1}.$$

Multiplied by $w_t/(\rho+\tau)$, these are the optimality conditions in eq. (11) of the main text, with the expected profits of a newly captured product line from eq. (6).

I now prove that $x^h > x^\ell$. By eq. (2), a high-type firm charges a higher markup than a low-type firm on a captured product line whichever type it displaces: λ against $\lambda\varphi^\ell/\varphi^h$ when the displaced firm is of the high type, and $\lambda\varphi^h/\varphi^\ell$ against λ when it is of the low type. It therefore suffices to show that the expansion R&D rate increases in the markups a firm expects on captured lines, which I do by totally differentiating the optimality condition.

Write the optimality condition for expansion R&D as[31]

$$\frac{\zeta-1}{\psi_x}(x^h)^\zeta + S_t\frac{Y_t}{w_t}\left(1-\frac{1}{\mu^h}\right) + (1-S_t)\frac{Y_t}{w_t}\left(1-\frac{1}{\mu^\ell}\right) = (\rho+\tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta-1},$$

where μ^h and μ^ℓ denote the markup charged on a captured product line when the displaced firm is of the high and of the low type. Totally differentiating with respect to the two markups and the expansion R&D rate and dividing by $(x^h)^{\zeta-2} > 0$ gives

$$\frac{Y_t}{w_t}\frac{1}{(x^h)^{\zeta-2}}\left[S\frac{d\mu^h}{(\mu^h)^2} + (1-S)\frac{d\mu^\ell}{(\mu^\ell)^2}\right] = \frac{\zeta(\zeta-1)}{\psi_x}\left(\rho+\tau-x^h\right)dx^h.$$

The left-hand side is positive for any increase in the markups, $d\mu^h, d\mu^\ell \geq 0$ and not both zero, and so is $\rho+\tau-x^h$, since a stationary firm-size distribution requires $\tau > x^h$. Hence $dx^h > 0$: the expansion R&D rate increases in the markups a firm expects, and the high type, whose markups are higher in both cases, expands faster, $x^h > x^\ell$.

Neutrality of the product-line weights The last item of Proposition 1 follows in three steps. First, the weights do not appear in the optimality conditions for expansion R&D above, nor in the free-entry condition (8): by the linearity of the value function, the expected value of a newly acquired product equals its value at

[31] This uses the optimality condition of the high-type firm but the one of the low type works equivalently.

the mean weight of one. Given Y_t/w_t, S, and τ, the choices x^h and x^ℓ and the value of entry are therefore the same for any distribution G. Second, the weights do not appear in the aggregate conditions that determine Y_t/w_t, S, τ, and z. The laws of motion of the distribution of quality and productivity gaps (Section D.4) involve only the rates at which products change hands, which are the same for all products. Production labor is $\frac{Y}{w}\int_0^1 \beta_i \mu_i^{-1} di$, which equals $\frac{Y}{w}\int_0^1 \mu_i^{-1} di$ because the weights have mean one and are independent of markups. This independence holds at time zero by assumption and is preserved thereafter, since neither incumbents nor entrants can direct innovation toward products with particular weights, so that the identity and the markup of the producer of a product evolve independently of its weight. Third, given x^h, x^ℓ, and τ, the number of products of a firm follows a birth-death process that does not involve the weights. Hence the equilibrium system, and with it x^h, x^ℓ, z, τ, S, g, and the distribution of the number of products, is identical for any G. The same argument applies period by period along a transition path. The weights enter only when the number of products is mapped into sales, $\text{Size}_f = Y \sum_{i \in N_f} \beta_i$.

D.3 Steady-state growth rate of aggregate variables

The growth rate of Q_t determines the growth rate of aggregate variables

$$g = \frac{\dot{Q}_t}{Q_t} = \frac{\partial \ln(Q_t)}{\partial t}.$$

Quality of a product in a given product line increases through expansion R&D or firm entry. For the growth rate of Q_t over a discrete time interval Δ, we have

$$\ln(Q_{t+\Delta}) = \int_0^1 \Big[(\Delta S x^h + \Delta(1-S)x^\ell + \Delta z)\ln(\lambda) + \ln(q_{t,i}) \Big] di$$

Every product is improved at the same rate, so weighting products by β_i leaves this expression unchanged. It follows that

$$\frac{\ln(Q_{t+\Delta}) - \ln(Q_t)}{\Delta} = \Big(S x^h + (1-S)x^\ell + z \Big) \ln(\lambda).$$

For $\Delta \to 0$, $g = \Big(S x^h + (1-S)x^\ell + z \Big) \ln(\lambda)$ as stated in Proposition 1.

D.4 Joint distribution of quality and productivity gaps

The distribution of quality and productivity gaps between incumbent and laggard firms determines the product markups, and thereby labor demand and the aggregates of Section D.1, as well as the share of product lines operated by each productivity type, the state variable of the firm's problem. This section derives it as a function of firm policies, so that the equilibrium distribution can be solved jointly with the policies.

The distribution of quality and productivity gaps across product lines is characterized by four differential equations. Expansion R&D rates depend only on a firm's type, x_t^h and x_t^ℓ, along the balanced growth path (Proposition 1) and along the transition (Supplemental Appendix F). The measure $\nu_t\left(\lambda, \frac{\varphi_f}{\varphi_{f'}}\right)$ of product lines in which an incumbent of productivity type φ_f is a λ quality step ahead of a $\varphi_{f'}$ type laggard follows

$$\begin{aligned}
\dot{\nu}_t\left(\lambda, \frac{\varphi^\ell}{\varphi^h}\right) &= (1-S_t)x_t^\ell S_t + z_t(1-p^h)S_t - \nu_t\left(\lambda, \frac{\varphi^\ell}{\varphi^h}\right)\tau_t \\
\dot{\nu}_t\left(\lambda, \frac{\varphi^\ell}{\varphi^\ell}\right) &= (1-S_t)x_t^\ell(1-S_t) + z_t(1-p^h)(1-S_t) - \nu_t\left(\lambda, \frac{\varphi^\ell}{\varphi^\ell}\right)\tau_t \\
\dot{\nu}_t\left(\lambda, \frac{\varphi^h}{\varphi^h}\right) &= S_t x_t^h S_t + z_t p^h S_t - \nu_t\left(\lambda, \frac{\varphi^h}{\varphi^h}\right)\tau_t \\
\dot{\nu}_t\left(\lambda, \frac{\varphi^h}{\varphi^\ell}\right) &= S_t x_t^h(1-S_t) + z_t p^h(1-S_t) - \nu_t\left(\lambda, \frac{\varphi^h}{\varphi^\ell}\right)\tau_t. \qquad \text{(A-7)}
\end{aligned}$$

Changes in ν_t are due to inflows and outflows arising from creative destruction. For example, in the first line, the measure of products with a low-type incumbent and high-type laggard $\nu_t\left(\lambda, \frac{\varphi^\ell}{\varphi^h}\right)$ increases due to low-type incumbents and entrants replacing high-type incumbents that occupy a share S_t of product lines, captured by $(1-S_t)x_t^\ell S_t + z_t(1-p^h)S_t$. The first term captures creative destruction by low-type incumbents and the second one creative destruction by low-type entrants. At the same time, the measure declines at rate τ_t as these firms themselves are subject to creative destruction. Adding the last two lines of (A-7) yields the law of motion of the share of product lines operated by high-productivity firms, $S_t = \nu_t\left(\lambda, \frac{\varphi^h}{\varphi^h}\right) + \nu_t\left(\lambda, \frac{\varphi^h}{\varphi^\ell}\right)$, stated in eq. (7).

To solve for $\nu_t\left(\lambda, \frac{\varphi^\ell}{\varphi^h}\right), \nu_t\left(\lambda, \frac{\varphi^\ell}{\varphi^\ell}\right), \nu_t\left(\lambda, \frac{\varphi^h}{\varphi^h}\right), \nu_t\left(\lambda, \frac{\varphi^h}{\varphi^\ell}\right)$ along the balanced growth

path, I set the differential equations in (A-7) equal to zero. Denote the solutions to these four equations by $S_{\varphi^\ell,\varphi^h}, S_{\varphi^\ell,\varphi^\ell}, S_{\varphi^h,\varphi^h}, S_{\varphi^h,\varphi^\ell}$.

The following uses the obtained results to solve for the aggregate labor income share, the misallocation measure and the aggregate markup. Since the product-line weights are independent of the gaps, the measures ν_t are also the expenditure shares of the four groups of products, and the expressions below hold for any distribution of the weights. The aggregate labor income share equals

$$\Lambda = \frac{wL_P}{Y} = \int_0^1 \mu_i^{-1} di = \frac{1}{\lambda} \sum_{k\in\{h,\ell\}} \sum_{n\in\{h,\ell\}} \frac{1}{\varphi^k/\varphi^n} S_{\varphi^k,\varphi^n}. \tag{A-8}$$

The TFP misallocation statistic $\mathcal{M}$ is given by

$$\mathcal{M} = \frac{\exp\left(\int_0^1 \ln \mu_i^{-1} di\right)}{\int_0^1 \mu_i^{-1} di} = \frac{1}{\lambda} \frac{\left(\frac{\varphi^h}{\varphi^\ell}\right)^{S_{\varphi^\ell,\varphi^h} - S_{\varphi^h,\varphi^\ell}}}{\Lambda}.$$

The aggregate markup (sales weighted across products) is given by

$$E[\mu] = \int_0^1 \mu_i di = \lambda \sum_{k\in\{h,\ell\}} \sum_{n\in\{h,\ell\}} \frac{\varphi^k}{\varphi^n} S_{\varphi^k,\varphi^n}. \tag{A-9}$$

The cost weighted aggregate markup equals $1/\Lambda$.

D.5 Firm dynamics

The following subsection derives the components of the size dispersion decomposition, in particular the age distribution and size dispersion conditional on age. A firm's sales are $\text{Size}_{f,t} = Y_t \sum_{i\in N_f} \beta_i$ (eq. (3)), where the weights of a surviving firm's products are independent draws from G. The distribution of log sales conditional on the number of products enters through its first two moments,

$$m_k(n) \equiv E\left[\left(\ln \sum_{j=1}^n \beta_j\right)^k\right], \quad k = 1, 2,$$

the moments of the log of a sum of n weights. With equal weights, $m_k(n) = (\ln n)^k$.

Mean size by age Mean log size conditional on age pools both productivity types with the type-share weights from eq. (14):

$$\mu(a_f) \equiv E\left[\ln \text{Size}_{f,t} \mid a_f\right] = s^h(a_f) \cdot \mu^h(a_f) + \left(1 - s^h(a_f)\right) \cdot \mu^\ell(a_f), \tag{A-10}$$

where $\mu^d(a_f) = E[\ln \text{Size}_{f,t} | a_f, \varphi_f = \varphi^d] = \ln Y_t + \sum_{n=1}^{\infty} \left(1 - \gamma^d(a_f)\right) \left(\gamma^d(a_f)\right)^{n-1} m_1(n)$ denotes the type-specific mean log size conditional on age. The *log of average size* conditional on age, the object shown in Figure 2 and targeted in the estimation, follows from the mean of the geometric distribution of product counts (eq. (15)):

$$\ln E\left[\text{Size}_{f,t} \mid a_f\right] - \ln E\left[\text{Size}_{f,t} \mid 0\right] = \ln \left[\frac{s^h(a_f)}{1 - \gamma^h(a_f)} + \frac{1 - s^h(a_f)}{1 - \gamma^\ell(a_f)}\right].$$

Age distribution Since each entrant independently draws a high-type productivity with probability p^h and a low-type productivity with probability $1 - p^h$, the unconditional survival probability to age a_f is

$$\chi(a_f) = p^h \chi^h(a_f) + (1 - p^h)\chi^\ell(a_f), \tag{A-11}$$

where χ^h and χ^ℓ are defined in eq. (13). In the numerical implementation of the decomposition, continuous age is approximated on a fine grid with step size Δa and midpoints $\{a_k\}$. The discrete probability mass assigned to each age bin is

$$\omega(a_k) = \frac{p^h \chi^h(a_k) + (1 - p^h)\chi^\ell(a_k)}{\sum_j \left[p^h \chi^h(a_j) + (1 - p^h)\chi^\ell(a_j)\right]}.$$

Size dispersion conditional on age Within any age cohort, log-size dispersion arises from three sources: the allocation of firms across productivity types, the stochastic accumulation of product lines, and the weights of these product lines. The number of product lines of a surviving firm of type d at age a_f follows a geometric distribution with parameter $\gamma^d(a_f)$ (see eq. (15)):

$$P(n_f = n \mid a_f,\, \varphi_f = \varphi^d) = \left(1 - \gamma^d(a_f)\right) \left(\gamma^d(a_f)\right)^{n-1}, \quad n = 1, 2, \ldots$$

Pooling both types with weight $s^h(a_f)$ from eq. (14), the mixture distribution of product count conditional on age is

$$P(n_f = n \mid a_f) = s^h(a_f) \cdot \big(1-\gamma^h(a_f)\big)\big(\gamma^h(a_f)\big)^{n-1} + \big(1-s^h(a_f)\big) \cdot \big(1-\gamma^\ell(a_f)\big)\big(\gamma^\ell(a_f)\big)^{n-1}.$$

The variance of log sales conditional on age a_f is therefore

$$\mathrm{Var}(\ln \mathrm{Size}_{f,t} \mid a_f) = \sum_{n=1}^{\infty} P(n_f = n \mid a_f) \cdot m_2(n) - \left(\sum_{n=1}^{\infty} P(n_f = n \mid a_f) \cdot m_1(n)\right)^2. \tag{A-12}$$

With equal weights, it reduces to the variance of the log number of products, $\sum_{n=1}^{\infty} P(n_f = n \mid a_f) \cdot (\ln n - \bar{\mu}(a_f))^2$ with $\bar{\mu}(a_f) = \sum_{n=1}^{\infty} \ln(n) \cdot P(n_f = n \mid a_f)$. The moments $m_1(n)$ and $m_2(n)$ have no closed form for Pareto weights. I compute them by simulation, from the partial sums of one million simulated sequences of weights for n up to 800, and hold the draws fixed across parameter values. The percentiles of log sales by age used in Table 4 come from the same simulated sequences: for each n, I tabulate the distribution of the log partial sum, and the percentile of log sales in an age group is obtained by inverting the mixture of these distributions over $P(n_f = n \mid a_f)$.

Implementation with age groups Empirically, firm age is observed at annual frequency. For firms of integer age a, i.e., $a_f \in [a, a+1)$, the law of total variance applied across exact continuous ages within the bin gives the within-bin size variance

$$V_a = \underbrace{\sum_{k:\, a_k \in [a,a+1)} \frac{\omega(a_k)}{p_a} \cdot \mathrm{Var}(\ln \mathrm{Size}_{f,t} \mid a_k)}_{\text{avg. within-exact-age variance}} + \underbrace{\sum_{k:\, a_k \in [a,a+1)} \frac{\omega(a_k)}{p_a} \cdot (\mu(a_k) - \bar{\mu}_a)^2}_{\text{variance of mean size across exact ages}}, \tag{A-13}$$

where $p_a = \sum_{k:\, a_k \in [a,a+1)} \omega(a_k)$ is the mass of firms in age bin a, $\mu(a_k)$ is the mean log size at fine-grid midpoint a_k from eq. (A-10), and $\bar{\mu}_a = \sum_{k:\, a_k \in [a,a+1)} (\omega(a_k)/p_a) \cdot \mu(a_k)$ is the within-bin conditional mean. The same formula, with $[a, a+1)$ replaced by the group's age range, gives the variance within an age group such as 6–10 or 21 and above, the objects of Section 6. The total cross-sectional variance then decomposes as

$$\mathrm{Var}(\ln \mathrm{Size}) = \underbrace{\sum_a p_a \cdot V_a}_{\text{within-age}} + \underbrace{\sum_a p_a \cdot (\bar{\mu}_a - \bar{\mu})^2}_{\text{between-age}},$$

where $\bar{\mu} = \sum_a p_a \bar{\mu}_a$ is the grand mean log size.

E Sensitivity to the entrant composition

The share of high-type entrants p^h is set to one half in the main text. This appendix re-estimates the new balanced growth path of Section 6 for alternative values. Two properties of the model delimit the exercise. First, p^h leaves the initial balanced growth path unchanged. With a productivity gap of unity the two types are identical, so the estimates of ψ_x, ψ_z, λ, and α in Table 3, and every moment of the initial balanced growth path, hold for any p^h. The entrant composition therefore enters only through the new balanced growth path, and the exercise re-estimates the productivity gap alone, to the same two targets and with all other parameters at their initial values, exactly as in the main text.

Second, the targeted rise in old-age dispersion restricts the values of p^h the model admits. In the model, the rise in size dispersion among old firms is the between-type variance of product counts among old survivors. It requires both types to be present in old cohorts and vanishes when either type dominates. For p^h at or below 0.3, the largest rise in dispersion among firms aged 21 and above that is attainable before entry converges to zero and the balanced growth path ceases to exist falls short of the targeted 0.2 log points (0.15 at $p^h = 0.2$ and, just short, 0.197 at $p^h = 0.3$). For $p^h = 0.9$, old cohorts consist almost entirely of high-type firms, and the attainable rise is 0.10 log points when the gap reaches the step size λ, at which point the markup of a low-type firm facing a high-type competitor, $\lambda\varphi^\ell/\varphi^h$, falls to one. This leaves p^h between 0.4 and 0.8, the range over which the estimate of the productivity gap is interior to the feasible set.

Table A-2 reports the results over this range. The results of the paper do not depend on the entrant composition. The rise in old-age dispersion is matched at every value of p^h and the young-firm share comes within one percentage point of its target; dispersion among entrants stays flat, as it does in the data; and aggregate TFP growth falls by 0.10 to 0.11 percentage points in every case. What depends on p^h is the inferred size of the rise in the productivity gap, which increases with the share of high-type entrants, from 2.3% at $p^h = 0.4$ to 7.9% at $p^h = 0.8$: the more entrants are born high type, the wider the gap between the two types must be to generate a given rise in dispersion among old firms. The quantitative rise in the gap is thus the one object that rests on the entrant composition; the age signature of the rise in size dispersion and the slowdown of growth do not.

Table A-2: New balanced growth path for alternative entrant compositions p^h

	Data	Share of high-type entrants p^h				
		0.4	0.5	0.6	0.7	0.8
Targeted						
$\Delta\sigma$ (ln Size|Age=21+)	0.200	0.208	0.208	0.207	0.204	0.200
Young-firm share in %	10.4	11.4	11.4	11.2	10.9	10.4
Untargeted						
$\Delta\sigma$ (ln Size|Age=0)	≈ 0	0.000	0.000	0.000	0.000	0.000
Δ TFP growth in pp	-0.43	-0.10	-0.10	-0.10	-0.10	-0.11
Implied parameter						
Productivity gap φ^h/φ^ℓ		1.023	1.027	1.034	1.047	1.079
Largest attainable $\Delta\sigma$ (ln Size|Age=21+)		0.27	0.33	0.34	0.33	0.29

Notes: Each column re-estimates the productivity gap φ^h/φ^ℓ to the two targeted moments (the rise in the standard deviation of log sales among firms aged 21 and above relative to the initial balanced growth path and the young-firm share) for the indicated share of high-type entrants, holding all other parameters at their initial-balanced-growth-path values, which do not depend on p^h. The column for $p^h = 0.5$ is the first productivity-gap column of Table 4 (SD target). $\Delta\sigma$ (ln Size|Age=0) is the change in the standard deviation of log sales among entrants (age zero), flat in the data. Δ TFP growth is the change in aggregate productivity growth across balanced growth paths, in percentage points, relative to its initial value of 1.5%; in the data, average TFP growth over 2010–2015 stood at 1.07% (FRED). The last row reports the largest rise in old-age dispersion the model can generate at the given p^h over the feasible range of the gap; it falls to 0.15, 0.197, and 0.10 log points at $p^h = 0.2$, 0.3, and 0.9, respectively, short of the targeted 0.2 log points.

F Computation of transition dynamics

In this section, I lay out the numerical procedure to solve for the transition path. Since time is continuous, I solve a discretized version of the model where the solution converges to the one in continuous time for small enough time intervals. As shown in Supplemental Appendix D.2, value functions are additive across product lines and linear in the weight of each product, and the product-line weights do not affect firm policies or aggregate outcomes. Therefore, I solve the problem of two representative one-product firms with a unit weight ($\beta_i = 1$): one of the high-productivity type and one of the low-productivity type.

I normalize the value function by the wage w_t to obtain a stationary problem. The value function for the high-type firm (in discrete time) reads

$$
\begin{aligned}
&\frac{V_t^h(1,\mu_i,S_t)}{w_t} = \frac{Y_t}{w_t}\left(1-\frac{1}{\mu_i}\right)dt \\
&- \tau_t \exp(-r_{t+dt}dt)\frac{V_{t+dt}^h(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}dt \\
&+ \max_{x_t^h}\left\{x_t^h \exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V_{t+dt}^h(1,\lambda,S_{t+dt})}{w_{t+dt}} + (1-S_{t+dt})\frac{V_{t+dt}^h(1,\lambda\frac{\varphi^h}{\varphi^\ell},S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_x}(x_t^h)^\zeta dt\right\} \\
&+ \exp(-r_{t+dt}dt)\frac{V_{t+dt}^h(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}.
\end{aligned}
$$

The value function for the low-type firm reads

$$
\begin{aligned}
&\frac{V_t^\ell(1,\mu_i,S_t)}{w_t} = \frac{Y_t}{w_t}\left(1-\frac{1}{\mu_i}\right)dt \\
&- \tau_t \exp(-r_{t+dt}dt)\frac{V_{t+dt}^\ell(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}dt \\
&+ \max_{x_t^\ell}\left\{x_t^\ell \exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V_{t+dt}^\ell(1,\lambda\frac{\varphi^\ell}{\varphi^h},S_{t+dt})}{w_{t+dt}} + (1-S_{t+dt})\frac{V_{t+dt}^\ell(1,\lambda,S_{t+dt})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t}dt - \frac{1}{\psi_x}(x_t^\ell)^\zeta dt\right\} \\
&+ \exp(-r_{t+dt}dt)\frac{V_{t+dt}^\ell(1,\mu_i,S_{t+dt})}{w_{t+dt}}\frac{w_{t+dt}}{w_t}.
\end{aligned}
$$

From this, one obtains the first-order conditions for the policy functions. For the optimal expansion R&D rate of the high-type firm x_t^h (again suppressing the dependence of the value function on S_t):

$$
\exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V_{t+dt}^h(1,\lambda)}{w_{t+dt}} + (1-S_{t+dt})\frac{V_{t+dt}^h(1,\lambda\frac{\varphi^h}{\varphi^\ell})}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t} = \frac{\zeta}{\psi_x}(x_t^h)^{\zeta-1} \tag{A-14}
$$

and for the low-type firm x_t^ℓ:

$$
\exp(-r_{t+dt}dt)\left(S_{t+dt}\frac{V_{t+dt}^\ell(1,\lambda\frac{\varphi^\ell}{\varphi^h})}{w_{t+dt}} + (1-S_{t+dt})\frac{V_{t+dt}^\ell(1,\lambda)}{w_{t+dt}}\right)\frac{w_{t+dt}}{w_t} = \frac{\zeta}{\psi_x}(x_t^\ell)^{\zeta-1}. \tag{A-15}
$$

Both are independent of the markup μ_i.

Equations (F) to (A-15) characterize the firm problem in discrete time. These equations are supplemented by the law of motion for the distribution of productivity gaps, eq. (A-7) in discrete time, in which $\nu_{t+dt} - \nu_t$ equals dt times the right-hand side, and by a standard Euler equation

$$\frac{C_{t+dt}}{C_t} = \exp(-\rho dt)(1 + r_{t+dt}dt). \tag{A-16}$$

Further, the (static) free entry and labor market clearing conditions remain unchanged and are characterized in the main text by equations (8) and (9).

The algorithm to compute the transition path assumes that the initial and final balanced growth paths have been solved for, including the (stationary) distribution of productivity gaps. I choose $dt = 0.02$ and set the transition period to 300 years (T), after which I assume the economy has reached its new balanced growth path. I then compute the transition path as follows:

1. Guess a path of wage growth $\frac{w_{t+dt}}{w_t}$ and income to wage ratios Y_t/w_t over the transition (equal to their values in the final balanced growth path). Given these paths, the interest rate follows from the Euler equation (A-16), since $C_t = Y_t$ and output growth is the product of wage growth and the change in Y_t/w_t.

   (a) Guess a path for S_t over the transition (equal to its value in the final balanced growth path).

      i. Starting backwards in period T, solve for optimal policy functions in $T - dt$ using equations (A-14) and (A-15).

      ii. Solve for τ_{T-dt} that ensures that the free entry condition (8) holds.

      iii. Compute the value function in $T - dt$ using equations (F) and (F).

      iv. Iterate backwards until the first time period.

      v. Starting from the initial balanced growth path, simulate S_t forward using[32]

      $$S_{t+dt} = S_t + dt\left[S_t x_t^h(1 - S_t) - (1 - S_t)x_t^\ell S_t + z_t(p^h(1 - S_t) - (1 - p^h)S_t)\right],$$

      where z_t can be substituted out by equation (5).

   (b) Update the guess for S_t toward the path implied in step v, as a weighted average of the guessed and the implied path with weight 0.35 on the implied

[32] One could already simulate the entire distribution of productivity gaps forward here. However, for the inner loop, it is sufficient to iterate on S_t.

path, and go back to step i. Iterate until the guessed and the implied path for S_t converge.

2. Starting from the initial balanced growth path, simulate the distribution of productivity gaps forward using the discretized eq. (A-7).

3. Solve for the implied sequence of $\frac{Y_t}{w_t}$ from the labor market clearing condition.

4. Compute the sequence of quality growth using

$$\frac{Q_{t+dt}}{Q_t} = \exp\left(\left[S_{t+dt}x^h_{t+dt} + (1 - S_{t+dt})x^\ell_{t+dt} + z_{t+dt}\right] dt \ln(\lambda)\right).$$

5. Compute the sequence of aggregate productivity growth using

$$\frac{\Phi_{t+dt}}{\Phi_t} = \left(\frac{\varphi^h}{\varphi^\ell}\right)^{S_{t+dt}-S_t}.$$

6. Using the distribution of productivity gaps, compute the sequence of $\mathcal{M}_t$ defined in equation (4).

7. Compute the sequence of production labor L_{Pt} using equation (A-4).

8. Compute the sequence of aggregate output growth $\frac{Y_{t+dt}}{Y_t}$ using equation (4).

9. With the paths of aggregate output growth and $\frac{Y_t}{w_t}$, obtain the implied path of wage growth $\frac{w_{t+dt}}{w_t}$.

10. Update the guesses for $\frac{w_{t+dt}}{w_t}$ and $\frac{Y_t}{w_t}$ in the same way, with weight 0.9 on the implied paths, and go back to step (a). Iterate until the guessed and implied paths converge.

I declare convergence when the difference between the three guessed and implied sequences is less than $\epsilon = 10^{-5}$ in absolute terms. I annualize wage growth before applying the convergence threshold.